

Master Thesis

Common Ground: Tracking Human-Robot Interaction through Coarse Force-Plate Sensing

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Contents

Preface	v
Abstract	vii
Symbols	ix
1 Introduction	1
1.1 Context and motivation	1
1.2 Previous work and research gap	2
1.2.1 Environmental perception for indoor HRI	2
1.2.2 High-resolution pressure-sensing floors	2
1.2.3 Coarse load-sensing floors and multi-agent tracking	3
1.2.4 Research gap	4
1.3 Research question and scope	4
1.4 Approach and contributions	5
1.5 Experimental evaluation	5
1.6 Thesis outline	6
2 Methodology	7
2.1 Coordinate and notation conventions	8
2.2 Force-floor acquisition, calibration, and spatial representation	8
2.2.1 Force-plate hardware and measurement model	8
2.2.2 Load-cell calibration	9
2.2.3 Filtering and activation threshold	11
2.2.4 Common world frame and plate geometry	11
2.3 Global agent-tracking framework	13
2.3.1 Unassigned footprints and blob formation	13
2.3.2 Blob-track continuity and agent birth	13
2.3.3 Persistent agent state and prediction	14
2.3.4 Claims and connected components	14
2.3.5 Global fractional force assignment	16
2.3.6 Sub-grid position QP	23
2.3.7 Cross-frame force sustainment	29
2.3.8 Assignment application and persistent agent state	31
2.3.9 Force-weighted influence field	32
2.4 Post-tracking applications	35
2.4.1 Scene- and model-informed tracking	35
2.4.2 Partition-based robot pose fusion	36
2.4.3 Touch-contact dynamics	40
2.4.4 Carrying dynamics	40
2.5 Simulation through the force-floor representation	43
2.5.1 Digital force-floor simulation	43

2.5.2	Human simulation	44
2.5.3	Robot simulation	44
2.5.4	Simulated trajectories	44
3	Experimental Validation	45
3.1	MoCap reference environment	45
3.1.1	Plate registration	45
3.1.2	Rigid-body marker frames, projected CoM references, and reference weights	46
3.2	Single-plate CoP characterization	47
3.3	Solo and multi-agent tracking evaluation	50
3.3.1	Episode design and time synchronization	50
3.3.2	Solo CoP distributions	51
3.3.3	Episode-level evaluation metrics	53
3.3.4	Results	54
4	Conclusion	57
4.1	Discussion	57
4.2	Limitations	58
4.3	Future work	59
	Bibliography	63
A	Exploratory tracking and recognition experiments	65
A.1	Identity reassignment and switch recovery	65
A.2	Agent removal solver	66
A.3	Online fitting of local footprint distributions	67
A.4	Directed graph representation for interaction recognition	68

Preface

This master’s thesis was conducted during a research internship at GVLab, at The University of Tokyo, as part of the Robotics, Systems and Control program at ETH Zürich. The work was carried out in the context of GVLab’s research on human–robot interaction and biomechanics, using a modular force-sensing floor manufactured by Tec Gihan Co., Ltd. as the main experimental platform. It is exploratory in nature and specifically aims to demonstrate the use of Tec Gihan’s force-sensing floor as a complementary, affordable sensing modality for human–robot interaction.

The research was supervised at The University of Tokyo by Prof. Dr. Gentiane Venture and Prof. Dr. Vincent Hernandez and was presented at ETH Zürich under the supervision of Prof. Dr. Marco Hutter, with Chenhao Li serving as student supervisor.

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Abstract

Continuous knowledge of where people, robots, and objects are, and how forces are transferred among them, is useful for safe Human–Robot Interaction (HRI) analysis, assistive robotics, and non-intrusive monitoring. Cameras provide abundant geometric and semantic information, but are affected by occlusion, lighting, and privacy concerns. Wearable sensors depend on user collaboration and can interfere with comfort. This thesis investigates a complementary sensing modality that depends on neither: a coarse smart floor formed by $0.5\text{ m} \times 0.5\text{ m}$ force plates that report the magnitude and location of local vertical force through their Center of Pressure (CoP).

The main difficulty introduced by this sensor configuration is measurement superposition. When several contacts load the same plate, the sensor returns only a single aggregated force and first moment. The proposed real-time pipeline therefore tracks persistent agents by associating their footprint contributions across force, space, and time. A Labeled Multi-Bernoulli birth layer aggregates unassigned footprint blobs and initializes new agents, while two physics-constrained convex optimization stages resolve ambiguous measurements. The Force Share QP determines fractional force ownership under neutral-weight, reach, spatial, and temporal constraints. The Sub-grid Position QP then separates the corresponding contact locations while preserving the measured first moment and exploiting diagonal moments from the four-corner load cells. The baseline measurement model is agnostic to gait and form factor, allowing humans, robots, and objects to be tracked within the same framework.

The recovered physical state enables additional HRI functionality, including robot-pose correction, force-weighted influence fields, three-dimensional reconstruction of vertical touch contacts, and explicit object carrying dynamics. Optional scene information, robot models, odometry, and external pose observations can refine these estimates without replacing the force and moment measured by the floor.

The individual plates were characterized against a motion-capture reference, producing a mean CoP position error of 2.10 cm and within-location noise of 2.0–2.2 mm, indicating that the dominant uncertainty is a configuration-dependent spatial bias. Multi-agent tracking was evaluated over 82 one-minute episodes involving two adult human participants, a Pepper robot, a Vstone MecanumRover 3, and rigid objects. Relative to the solo baseline, the coordinate-wise CoP spread increased by 0.61–2.63 cm, the mean CoP-to-CoM offset of the rigid robots by 0.58–1.04 cm, and the terminal neutral-weight error by 1.21–3.95 N. These changes remain comparable to the intrinsic spatial uncertainty of the plates and small relative to their 0.5 m pitch. In three-agent scenes, the pipeline required 9.47 ms per update on average, below the 10 ms sensor period. The main unresolved failure mode occurs when the solver favors a lower-cost but incorrect explanation, causing an identity switch that the system cannot currently recover from. Overall, coarse vertical-force sensing provides useful real-time tracking and physical-interaction information for multiple agents, with degradation remaining close to the intrinsic sensing uncertainty, although not every identity or contact configuration can be uniquely resolved.

Symbols

Notation used repeatedly across the thesis is listed here; symbols local to one derivation are defined where they first appear. Scalars are italic, vectors are bold lowercase upright symbols, matrices are bold uppercase upright symbols, and coordinate frames are calligraphic. A hat denotes a normalized quantity, ^{sol} a solver result, ⁰ an agent’s learned neutral force, and ^{fix} a contribution fixed before the current solve. World-frame coordinates are used by default after plate registration.

Symbol	Meaning
$\mathcal{I}, \mathcal{S}_s$	inertial world frame and local frame of plate sensor s
$k, \Delta t_s$	force-floor frame index and sensor sampling period
$s, \ell \in \{1, 2, 3, 4\}$	force-plate sensor index and load-cell index
p, b, a, d	footprint, unassigned blob, and persistent-agent indices
$\mathcal{S}, \mathcal{P}_k, \mathcal{A}_k$	plate-sensor, active-footprint, and current-agent sets
$\theta_s, \mathcal{I} \mathbf{r} \mathcal{I} \mathcal{S}_s$	registered yaw and world-frame origin of plate sensor s
$\mathcal{B}_s, \mathcal{B}_p$	physical support of plate sensor s and source footprint p
$F_{p,k}, F_{p,\ell,k}$	total vertical force and load-cell forces of footprint p
$\mathbf{r}_{p,k}$	world-frame CoP of footprint p
$\mathbf{x}_{a,k}$	persistent state $[r^x, r^y, v^x, v^y, F, \dot{F}]^T$ of agent a
$\mathbf{r}_{a,k}, F_{a,k}$	tracked world-frame position and vertical force of agent a
F_a^0, N_a	learned neutral force and current assignment-force target of agent a
$C_{ap,k}, e_{p,k}$	claim-eligibility and floor-edge indicators
$X_{ap,k}, u_{p,k}$	assigned and unassigned fractions of source footprint p
$F_{ap,k}, \mathbf{y}_{ap,k}$	applied force share and resolved position of footprint p for agent a
$\mathbf{h}_{a,k}, \theta_{a,k}^{\text{MC}}$	MoCap marker-frame position and yaw of entity a
$\mathbf{o}_a, \mathbf{c}_{a,k}$	marker-frame-to-CoM offset and projected-CoM reference of entity a
Acronym	Meaning
CoM	Center of Mass
CoP	Center of Pressure
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
HRI	Human–Robot Interaction
LMB	Labeled Multi-Bernoulli
MoCap	Motion Capture
QP	Quadratic Program
RMSE	Root-Mean-Square Error
URDF	Unified Robot Description Format

Chapter 1

Introduction

1.1 Context and motivation

Robots are increasingly expected to operate alongside people in shared indoor environments rather than inside isolated work cells. Domestic service, clinical support, education, hospitality, logistics, and collaborative research all require robots to navigate around people, coordinate tasks with them, and, in some cases, exchange physical support or objects. These environments are dynamic: people and robots enter and leave, paths cross, objects move, and contacts occur outside predefined interaction areas. Effective human–robot interaction (HRI) therefore requires not only detecting occupants, but also tracking their identities and locations and recognizing when forces are transferred among people, robots, objects, and the environment. Perceiving nearby people is consequently a central problem in human-aware robot navigation and interaction [1].

Most indoor HRI systems obtain this information from sensors mounted on the robot or distributed throughout the environment. Cameras and depth sensors provide detailed geometric and semantic information, but depend on viewpoint, lighting, and line of sight. Close encounters and crowded scenes can cause self-occlusion and mutual occlusion precisely when reliable estimates are most important. Cameras also collect identifiable visual information, which can make them unsuitable for private or continuously monitored spaces. Wearable inertial and radio-frequency systems avoid some of these limitations, but require each user to carry, charge, and correctly associate a device. Neither approach directly measures the support forces through which many physical interactions are expressed.

An instrumented floor provides a complementary, environment-centered view of the same scene. People, ground-supported robots, and resting objects transmit load to the floor during normal indoor activity. Measuring this load is unaffected by lighting or visual occlusion, requires no worn device, and directly captures physical support and load transfer. This information can support human-aware navigation, interaction analysis, robot–object and human–robot handover reasoning, and contact-aware safety monitoring. Aging in place is one important application: falls are a major health risk for older adults, while continuous in-home observation raises concerns about privacy, installation, maintenance, and acceptance [2, 3].

To provide an affordable and easily installed method for monitoring indoor activity, Tec Gihan Co., Ltd. [4] developed a prototype $0.5\text{ m} \times 0.5\text{ m}$ force-sensing plate with four vertical load cells near its corners. Its rigid construction supports repeated dynamic loading, while the modular design allows the sensing area to be rearranged without permanently modifying the room. From the four measured loads, the total vertical force and planar center of pressure (CoP) can be recovered, as in conventional

ground-reaction-force analysis.

The large size of the plates increases coverage and reduces the number of sensors required, but also limits spatial resolution. Their pitch is comparable to a human stance or the full support area of a small mobile robot. When several feet, wheels, or objects load the same plate, the hardware reports only their combined force and first moment. Their individual forces and contact locations are therefore not directly observable. The central objective is to determine whether calibration, temporal tracking, and physically constrained inference can recover useful HRI information despite this limited resolution.

The problem is therefore not simply localization on a known grid. It requires separating shared force measurements into physically consistent agent contributions while identities, contact patterns, and the number of agents change over time. This thesis investigates how much interaction-relevant information can be recovered in software from these coarse measurements without increasing sensor density or assuming a fixed human gait, leg count, or robot footprint.

1.2 Previous work and research gap

1.2.1 Environmental perception for indoor HRI

Indoor HRI perception commonly relies on robot-mounted vision, depth, and ranging sensors, sometimes combined with fixed cameras or other ambient sensors. These modalities provide geometry, appearance, and semantic information and form the basis of human-aware navigation in shared spaces. Their limitations depend on how they are installed: onboard sensors have a moving and often restricted viewpoint, while fixed sensors cover only their installed field of view. Both can lose people or objects through occlusion. Combining several sensing modalities can reduce these problems, but introduces additional registration, synchronization, and data-association requirements.

A floor sensor instead provides a persistent, world-fixed measurement plane and observes a different physical quantity. It cannot recover appearance or full-body pose on its own, but it can continue measuring support when visual observations are occluded and can indicate whether a visible event is accompanied by a transfer of load.

This distinction is particularly relevant to physical HRI. Vision can provide geometric and semantic cues that a person touched a robot or lifted an object, while force exchanged with the ground provides independent information about the magnitude and dynamics of that interaction without requiring additional robot-mounted sensors or wearable devices. Floor sensing can therefore act as an additional physical sensing channel for HRI.

1.2.2 High-resolution pressure-sensing floors

Early and recent smart floors show that dense pressure measurements can contain detailed information about occupants. The high-resolution floor of Morishita et al. used 4096 pressure switches in each $0.5\text{ m} \times 0.5\text{ m}$ unit and 65,536 switches over a $2\text{ m} \times 2\text{ m}$ area. Its approximately 7 mm spatial resolution allowed a human and a wheeled cart to be separated from their pressure patterns [5]. Large-area analog systems have similarly produced pressure images for movement-based interaction, although their design must balance spatial resolution and coverage against frame rate, latency, data transmission, compression, and calibration [6]. More recent modular systems continue to address room-scale construction and reconfiguration [7].

Dense tactile carpets have enabled richer forms of human-state estimation. Pressure images have been used for three-dimensional human pose and mesh reconstruction [8, 9], while pressure has also been combined with vision and motion capture to improve foot-contact and motion reconstruction [10, 11]. Recent work extends pressure-image processing to multi-person tracking and human mesh recovery [12]. Together, these studies show that human-floor contact contains useful information and that pressure sensing can complement, or in constrained tasks replace, visual input.

High spatial resolution, however, comes with increased sensing and readout complexity. Large areas require many sensing elements and increasing amounts of data to maintain real-time operation. Some dense floors provide only binary contact, while analog pressure carpets require consistent calibration across sensing elements if force magnitude is to be interpreted. Their reconstruction methods also often assume dense, image-like measurements and human-specific training data or body priors. These assumptions suit detailed human-pose reconstruction, but transfer less directly to HRI scenes containing humans, different robot forms, and passive objects. More importantly, a method based on resolved pressure shapes cannot reconstruct shapes that were never measured by a coarse sensor.

The approach studied here is deliberately sparse. Four corner loads summarize all contact on a plate through its total force and first moment. This reduces sensing and readout complexity and allows reconfigurable coverage with existing hardware, but removes the pressure image used by dense-floor segmentation and pose-reconstruction methods. Missing information must instead be recovered, where possible, from physical constraints and temporal continuity.

1.2.3 Coarse load-sensing floors and multi-agent tracking

Coarse load-sensing floors have been studied as unobtrusive sensors for ambient intelligence and indoor localization. Andries et al. localized and recognized humans, robots, and objects on a low-resolution tiled floor by segmenting connected load regions, tracking them, and reasoning about their possible contents using weight [13]. The method explicitly addressed crossings and weight ambiguity, while related work used probabilistic processing to improve robot localization from noisy, low-resolution floor measurements [14]. These studies are among the closest precedents for heterogeneous force-floor tracking because they are not limited to human pose reconstruction.

Their representation, however, remains centered on observed regions or blobs. When agents are separated, a connected region can often be associated with one entity. When they approach each other, their support regions merge and a single measurement may contain contributions from several entities. Maintaining possible blob identities preserves some tracking information, but does not determine how much of the measured force belongs to each agent or where each contribution acts within the plate. These quantities are needed to reason about physical interaction, and any proposed separation must remain consistent with the force and first moment measured by the hardware.

General multi-target tracking methods address changing target number, missed detections, clutter, and identity. The labeled multi-Bernoulli (LMB) filter provides a tractable labeled random-finite-set representation of target existence and state [15]. Related labeled random-finite-set methods also consider merged measurements [16, 17]. These methods provide a principled basis for persistence and association, but a generic merged-measurement model does not capture the mechanics of a tiled force floor. In particular, it does not guarantee conservation of each plate’s measured force and first moment, constrain reconstructed contacts to the physical plate, or use the four corner loads to distinguish between moment-equivalent solutions.

1.2.4 Research gap

The literature therefore leaves a gap between two established approaches. Dense pressure floors recover detailed contact patterns, but require high sensor density and are often specialized to human activity or pose. Coarse load-sensing floors provide larger or reconfigurable coverage with fewer measurements and can localize heterogeneous entities, but existing methods do not reconstruct physically conserved, agent-specific contributions when measurements are shared.

What is missing is a real-time method that operates directly on the sparse measurements available from four-corner force plates and simultaneously:

- discovers and maintains an initially unknown number of persistent agents;
- accommodates human, robot, and object support patterns without assuming a fixed gait or form factor;
- assigns shared plate measurements fractionally while retaining explicit unexplained force;
- conserves measured force and first moment during sub-grid contact reconstruction; and
- exposes support and load-transfer information for downstream HRI reasoning.

This thesis addresses this gap by combining probabilistic birth tracking with physically constrained optimization. Rather than attempting to recover information that is absent from the measurements, the method preserves the quantities measured by the sensors and makes the role of temporal, geometric, and model-based priors explicit.

1.3 Research question and scope

This thesis asks:

To what extent can a coarse grid of uniaxial force plates provide gait- and form-factor-agnostic multi-agent tracking and interaction-relevant force attribution in shared indoor HRI environments?

The question is framed in terms of extent rather than complete observability. The desired output is a set of persistent agent identities, attributed vertical forces, and support locations that can inform physical-interaction reasoning. The objective is not to reconstruct full-body pose, identify people biometrically, or guarantee identity through every ambiguous configuration. Different contact arrangements can produce the same coarse vertical-force measurement, so a force-only system cannot resolve every case.

Gait- and form-factor-agnostic refers to the core birth and assignment process. It does not require a predefined gait class, number of legs, or robot support shape. Optional geometry and dynamics models can constrain a known robot after identification, but do not change the force and moment measured by the floor. The system is therefore evaluated as a complementary environmental sensor for HRI rather than as a replacement for onboard or ambient perception.

1.4 Approach and contributions

The proposed pipeline converts each calibrated plate sample at 100 Hz into a world-frame *footprint* containing its total force, CoP, four corner loads, and physical support region. Unassigned footprints are clustered with DBSCAN [18]. An LMB birth layer tracks the resulting blobs over time and creates a persistent agent only after sufficient existence and force evidence [15]. The LMB is used only for agent birth. Once created, agents are maintained through motion prediction, footprint claims, constrained measurement assignment, and linear Kalman filtering [19]. The principal contributions are:

- a tracking lifecycle for heterogeneous, initially unknown agents;
- a claim-graph decomposition and convex Force Share quadratic program that assigns each contested footprint fractionally to eligible agents or to an explicit residual while conserving measured force;
- a convex Sub-grid Position quadratic program that reconstructs one contact position for each retained force share within the source plate, conserves the measured first moment exactly, and uses diagonal moments from the four corner loads to distinguish otherwise equivalent solutions;
- a compact force-weighted influence field representing the extent, correlation, and asymmetry of each agent’s current support;
- model-informed extensions for robot contact fitting, vertical touch reconstruction, and object pickup, carrying, and release events;
- a hardware-matched simulation interface; and
- an experimental evaluation that identifies the intrinsic plate error and isolates the additional error caused by multi-agent measurement sharing and identity maintenance.

The two optimization stages address measurement superposition directly. The Force Share program determines how much of each plate measurement belongs to each agent, while the Sub-grid Position program determines where those contributions act. Keeping these decisions separate preserves convexity, allows exact physical accounting, and distinguishes information measured by the hardware from information selected through priors.

The same plate-level interface is reproduced in simulation. Human contacts are generated from a public walking-biomechanics dataset [20], robot contacts are obtained from Isaac Lab [21], and multi-agent trajectories are generated using Optimal Reciprocal Collision Avoidance [22]. Simulation can therefore vary the size and density of the environment without changing the tracker input.

1.5 Experimental evaluation

The evaluation first separates sensor error from tracking error. All plate poses are registered within an optical motion-capture volume, while rigid-body marker offsets and reference weights are measured independently using a six-degree-of-freedom Bertec force plate [23]. Static tests then compare each plate’s CoP with the projected center of mass of a rigid reference object. The mean position error is 2.10 cm and the RMSE is 2.32 cm, while the 2.0–2.2 mm within-location variation shows that the dominant error is configuration-dependent spatial bias rather than short-term measurement noise.

End-to-end tracking is evaluated over 82 one-minute episodes: 41 solo baselines and 41 multi-agent recordings involving Pepper [24], MecanumRover 3 [25], two adult participants, and rigid objects. Comparing the multi-agent episodes with matched solo baselines isolates the additional error caused by shared footprints and persistent identity tracking. Across agent types and coordinate axes, the CoP spread increases by 0.61–2.63 cm. For the rigid robots, the mean CoP-to-CoM offset increases by 0.58–1.04 cm, while the terminal neutral-weight error increases by 1.21–3.95 N.

For scenes with three simultaneous agents, mean processing times are 3.92 ms for the force solver, 2.70 ms for the position solver, and 2.85 ms for the remainder of the pipeline on a laptop CPU. The resulting mean total of 9.47 ms remains below the 10 ms sensor period, allowing real-time operation at 100 Hz.

These results show that the additional error introduced by multi-agent tracking remains small relative to the 0.5 m plate pitch and is comparable to the intrinsic spatial error of the plates, while maintaining computational efficiency in commercial hardware. The main unresolved failure mode is identity ambiguity: when several physically plausible assignments explain the same measurements, the solver may select a lower-cost solution that exchanges agent identities.

1.6 Thesis outline

The remainder of the thesis first presents the force-floor measurement model and the complete tracking, optimization, and post-tracking pipeline. It then describes the experimental setup and reference-frame registration, characterizes the individual plates, and evaluates the system in solo and multi-agent conditions. The final chapter discusses the limits of force-only observability, summarizes the implications for indoor HRI, and identifies directions for multimodal identity resolution, denser-scene tracking, and computational optimization.

Chapter 2

Methodology

This chapter develops the force-floor tracking method from the measured plate loads to persistent agent states and directed interaction recognition. After the force measurements are calibrated, filtered, and transformed into a common world frame, each active plate is represented as a footprint containing vertical force, center of pressure (CoP), and physical plate support.

Tracking is introduced first in its uninformed form, using only these floor measurements. Footprints without an existing owner are grouped spatially into blobs, and a labeled multi-Bernoulli (LMB) [15] birth layer aggregates those blobs across time. A sufficiently persistent blob track initializes a new agent. Persistent agents then predict their state, claim physically reachable footprints, accept unambiguous claims directly, and resolve ambiguous or shared plates through force-allocation and sub-grid position optimization. The resulting footprint assignments update the agent state, while any remaining unassigned force returns to the blob-birth path. Scene information, known physical properties, robot models, and external pose observations are introduced later as optional information that can constrain this same measurement-driven pipeline.

A *footprint* is the active force and CoP measurement associated with one force plate. A *blob* is a spatial grouping of unassigned footprints used to accumulate evidence for a new agent. An *agent* is a persistent tracked entity initialized from a confirmed blob track. After agents exist, a *cluster* denotes a current-frame spatial grouping of agents and footprints used for local reasoning and disambiguation.

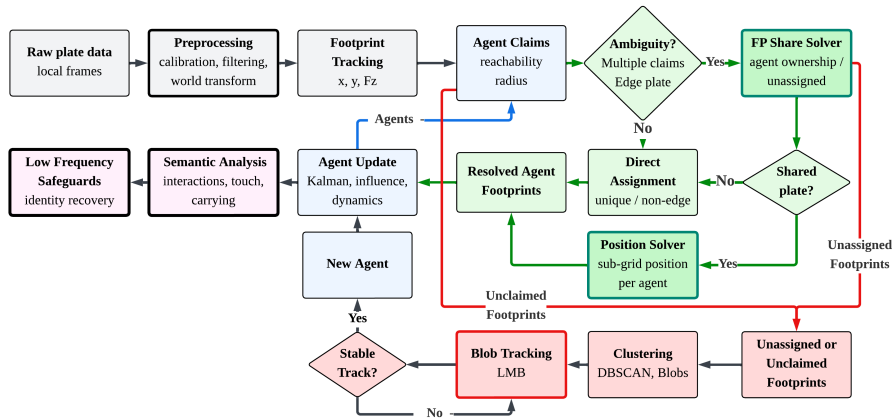


Figure 2.1: End-to-end methodology overview.

2.1 Coordinate and notation conventions

The notation follows the nomenclature used throughout this thesis. Scalars are italic. Vectors are written as bold lowercase upright symbols, matrices as bold uppercase upright symbols, and coordinate frames as calligraphic letters. The fixed inertial frame $\mathcal{I}$ is used as the world frame. When the frame of expression must be stated explicitly, it is written as a lower-left index; for example, ${}_Q\mathbf{r}_{QP}$ denotes the position vector from the origin Q of frame Q to point P , expressed in Q . Explicit frame notation is retained when introducing sensor-local quantities, performing a frame transformation, or combining quantities expressed in different frames. Otherwise, the active frame is stated in the surrounding text and suppressed from the symbol. In particular, after force-plate measurements have been transformed into the world frame, world coordinates are the default and positions are written compactly as $\mathbf{r}_p$, $\mathbf{r}_a$, and so forth. The orientation of frame $\mathcal{W}$ with respect to frame Q is denoted by ${}_Q\mathbf{R}_\mathcal{W} \in SO(3)$. Because the force floor is planar, θ denotes yaw and $\mathbf{R}(\theta) \in SO(2)$ denotes the corresponding planar rotation matrix.

Indices are fixed across the chapter: s denotes a force-plate sensor, $\ell \in \{1, 2, 3, 4\}$ one of its four load cells, p a footprint, b a blob, a and d agents, c a calibration record, i a sub-grid participant, m a model contact, and k a discrete sensor frame. Additional local counting indices are defined where they appear. The force-plate sensor, active-footprint, and current-agent sets are S , $\mathcal{P}_k$, and $\mathcal{A}_k$, respectively. The indexed calligraphic symbol $\mathcal{S}_s$ denotes the local coordinate frame attached to plate sensor s ; it is distinct from the unindexed sensor set S . All measured forces are vertical, along z , so force variables omit an additional directional frame index unless required for clarity; F_a^0 denotes the neutral vertical force of agent a . Later sections use X_{ap} for a scalar fractional ownership share and u_p for its scalar unassigned remainder. A hat denotes a normalized quantity, sol identifies a solver result, and a superscript star is reserved for a desired value.

2.2 Force-floor acquisition, calibration, and spatial representation

2.2.1 Force-plate hardware and measurement model

The sensing floor consists of 15 instrumented force plates arranged on a planar grid. Each plate is a $0.5\text{ m} \times 0.5\text{ m}$ square supported by four independent load cells located near its corners. Four plates are connected to each acquisition interface, and four interfaces are sampled concurrently, leaving one unused position in the resulting 4×4 hardware arrangement.

All sensing channels are sampled at the nominal frequency $f_s = 100\text{ Hz}$, corresponding to the sampling period $\Delta t_s = 1/f_s = 0.01\text{ s}$.

Let $s \in S$ denote a force-plate sensor, $\ell \in \{1, 2, 3, 4\}$ one of its four load cells, and k the sample index. The raw signed sensor count $n_{s\ell,k}^{\text{ADC}}$ is first corrected using the stored zero offset $o_{s\ell}$ and device scale factor $\kappa_{s\ell}^{\text{ADC}}$, provided by the acquisition hardware:

$$F_{s\ell,k}^{\text{raw}} = \kappa_{s\ell}^{\text{ADC}} (n_{s\ell,k}^{\text{ADC}} - o_{s\ell}).$$

The channel-specific calibration introduced in section 2.2.2 is then applied independently to each load cell:

$$F_{s\ell,k}^{\text{cal}} = a_{s\ell} (F_{s\ell,k}^{\text{raw}})^2 + b_{s\ell} F_{s\ell,k}^{\text{raw}}. \quad (2.1)$$

Applying the calibration before combining the load-cell channels preserves the intra-plate load distribution. This is required not only to recover the total normal force, but also to estimate the location at which the resultant vertical load acts.

The total vertical force measured by plate sensor s is

$$F_{s,k}^{\text{cal}} = \sum_{\ell=1}^4 F_{s\ell,k}^{\text{cal}}.$$

Let $\mathcal{S}_s$ denote the local coordinate frame of plate sensor s and d_s its calibrated load-cell lever arm. The local center of pressure (CoP) is obtained from the relative loading of the four load cells as

$$\mathcal{S}_s r_{s,k}^{x,\text{cal}} = -d_s \frac{F_{s2,k}^{\text{cal}} + F_{s3,k}^{\text{cal}} - F_{s1,k}^{\text{cal}} - F_{s4,k}^{\text{cal}}}{F_{s,k}^{\text{cal}}}, \quad (2.2)$$

$$\mathcal{S}_s r_{s,k}^{y,\text{cal}} = d_s \frac{F_{s1,k}^{\text{cal}} + F_{s2,k}^{\text{cal}} - F_{s3,k}^{\text{cal}} - F_{s4,k}^{\text{cal}}}{F_{s,k}^{\text{cal}}}. \quad (2.3)$$

The resulting planar CoP vector is

$$\mathcal{S}_s \mathbf{r}_{s,k}^{\text{cal}} = \begin{bmatrix} \mathcal{S}_s r_{s,k}^{x,\text{cal}} \\ \mathcal{S}_s r_{s,k}^{y,\text{cal}} \end{bmatrix}. \quad (2.4)$$

The division in eqs. (2.2) and (2.3) is evaluated only when $F_{s,k}^{\text{cal}} > 1 \text{ N}$ to prevent numerical instability when the plate is effectively unloaded.

Each plate measurement therefore retains its total force, four corner forces, and two CoP coordinates. Although each plate provides only one spatial sensing cell at the floor level, its four independent force measurements additionally encode the load distribution within the plate, as illustrated in fig. 2.2.

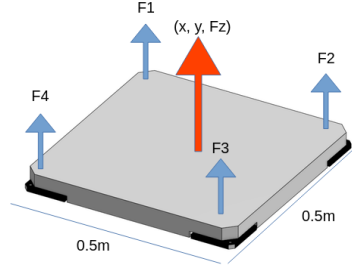


Figure 2.2: Force-plate hardware and load-cell geometry.

2.2.2 Load-cell calibration

To reduce sensor-to-sensor and load-cell response bias, the four load cells of each plate sensor are calibrated jointly from raw, unfiltered measurements before normal tracking operation. Known loads are placed at the center and four corner regions of each sensor, together with an unloaded measurement, and the corresponding raw load-cell forces are recorded. Using several application positions changes the distribution of the known total load among the four load cells and thereby provides information about their individual response characteristics.

For calibration record $r \in \{1, \dots, N_r\}$, let $\bar{F}_{s\ell,r}^{\text{raw}}$ denote the recorded mean raw force of load cell ℓ , and let m_r^{ref} be the applied reference mass. The corresponding reference total force is

$$F_{s,r}^{\text{ref}} = \begin{cases} 0, & \text{unloaded observation,} \\ m_r^{\text{ref}} g_0, & \text{loaded observation,} \end{cases} \quad g_0 = 9.80665 \text{ m s}^{-2}.$$

Each load cell is modeled by a zero-intercept quadratic response,

$$F_{s\ell} = a_{s\ell} (F_{s\ell}^{\text{raw}})^2 + b_{s\ell} F_{s\ell}^{\text{raw}},$$

where $a_{s\ell}$ and $b_{s\ell}$ are the calibration coefficients of load cell ℓ on sensor s . For calibration record r , the predicted total sensor force is therefore

$$F_{s,r}^{\text{pred}} = \sum_{\ell=1}^4 \left[a_{s\ell} (\bar{F}_{s\ell,r}^{\text{raw}})^2 + b_{s\ell} \bar{F}_{s\ell,r}^{\text{raw}} \right].$$

The coefficients are selected so that the predicted total force approaches the known applied force over all recorded calibration conditions. The calibration problem is

$$\min_{\{a_{s\ell}, b_{s\ell}\}_{\ell=1}^4} \sum_{r=1}^{N_r} \left[\sum_{\ell=1}^4 \left(a_{s\ell} (\bar{F}_{s\ell,r}^{\text{raw}})^2 + b_{s\ell} \bar{F}_{s\ell,r}^{\text{raw}} \right) - F_{s,r}^{\text{ref}} \right]^2. \quad (2.5)$$

Although the channel response is quadratic in the measured force, the unknown coefficients enter eq. (2.5) linearly. The problem can therefore be written as an ordinary linear least-squares system. For calibration record r , define

$$(\mathbf{h}_{s,r}^{\text{cal}})^{\text{T}} = [(\bar{F}_{s1,r}^{\text{raw}})^2 \quad \dots \quad (\bar{F}_{s4,r}^{\text{raw}})^2 \quad \bar{F}_{s1,r}^{\text{raw}} \quad \dots \quad \bar{F}_{s4,r}^{\text{raw}}].$$

For the N_r calibration records,

$$\mathbf{H}_s^{\text{cal}} = \begin{bmatrix} (\mathbf{h}_{s,1}^{\text{cal}})^{\text{T}} \\ \vdots \\ (\mathbf{h}_{s,N_r}^{\text{cal}})^{\text{T}} \end{bmatrix}, \quad \mathbf{f}_s^{\text{ref}} = [F_{s,1}^{\text{ref}} \quad \dots \quad F_{s,N_r}^{\text{ref}}]^{\text{T}}.$$

The eight calibration coefficients are collected as

$$\boldsymbol{\psi}_s^{\text{cal}} = [a_{s1} \quad \dots \quad a_{s4} \quad b_{s1} \quad \dots \quad b_{s4}]^{\text{T}}.$$

Provided that $\mathbf{H}_s^{\text{cal}}$ has full column rank, the least-squares solution is

$$\boldsymbol{\psi}_s^{\text{cal,sol}} = \arg \min_{\boldsymbol{\psi}_s^{\text{cal}}} \left\| \mathbf{H}_s^{\text{cal}} \boldsymbol{\psi}_s^{\text{cal}} - \mathbf{f}_s^{\text{ref}} \right\|_2^2 = (\mathbf{H}_s^{\text{cal}})^{\dagger} \mathbf{f}_s^{\text{ref}}.$$

The spatially varied calibration loads cause the quadratic and linear columns associated with the four load cells to vary independently across the recorded conditions. The resulting polynomial representation is stored per load cell and applied to the raw channels before the four calibrated forces are combined to obtain the sensor total force and CoP in eqs. (2.1) and (2.4). Filtering and activation thresholding are applied only after calibration.

2.2.3 Filtering and activation threshold

The calibrated force-plate measurements are available to the tracking system at $f_s = 100$ Hz. Before footprint construction, the four load-cell force channels are independently low-pass filtered with cutoff frequency $f_c = 5$ Hz, yielding $F_{s\ell,k}^{\text{filt}}$. Their sum defines $F_{s,k}^{\text{filt}}$, and the filtered local CoP $\mathcal{S}_s \mathbf{r}_{s,k}^{\text{filt}}$ is computed from those same filtered channels using eqs. (2.2) and (2.3). This preserves the force-moment consistency of each retained measurement.

Low-force measurements are rejected before entering the tracking pipeline to minimize the influence of noise introduced by disturbances in the environment. A plate is considered active only when its filtered total vertical force matches or exceeds the activation threshold $F_{\min} = 15$ N.

Due to manufacturing defects of the prototype, the sensors exhibit an unreliable CoP response when the measured force is below $F_{\min}^{\text{CoP}} = 40$ N. Hereafter, the CoP is considered reliable only when the measured force exceeds this threshold.

The measurement retained for tracking is therefore

$$(\mathcal{F}_{s,k}, \mathcal{S}_s \mathbf{r}_{s,k}) = \begin{cases} (F_{s,k}^{\text{filt}}, \mathcal{S}_s \mathbf{r}_{s,k}^{\text{filt}}), & F_{s,k}^{\text{filt}} \geq F_{\min}, \\ (0, \mathbf{0}), & F_{s,k}^{\text{filt}} < F_{\min}. \end{cases}$$

2.2.4 Common world frame and plate geometry

The measurements retained by section 2.2.3 are initially expressed independently in the local coordinate frame $\mathcal{S}_s$ of each plate. Tracking across several plates requires these local measurements to be represented in a single common world frame $\mathcal{I}$.

For plate sensor s , let $\mathcal{I} \mathbf{r}_{IS_s} \in \mathbb{R}^2$ denote the calibrated position of the plate origin in the world frame, and let θ_s denote its registered planar orientation in the floor layout. The methodology to obtain these values is discussed in section 3.1.1.

The corresponding planar rotation matrix is

$$\mathbf{R}(\theta_s) = \begin{bmatrix} \cos \theta_s & -\sin \theta_s \\ \sin \theta_s & \cos \theta_s \end{bmatrix},$$

and the homogeneous transformation from the plate-local frame to the world frame is

$$\mathcal{I} \mathbf{T}_{\mathcal{S}_s} = \begin{bmatrix} \mathbf{R}(\theta_s) & \mathcal{I} \mathbf{r}_{IS_s} \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

The local sensor coordinate convention has the opposite x direction from the geometric plate convention. This is corrected using $\mathbf{D}_{\text{CoP}} = \text{diag}(-1, 1)$, such that

$$\mathcal{S}_s \mathbf{r}_{S_s C_{s,k}}^{\text{grid}} = \mathbf{D}_{\text{CoP}} \mathcal{S}_s \mathbf{r}_{S_s C_{s,k}}.$$

The corresponding world-frame CoP is then

$$\begin{bmatrix} \mathcal{I} \mathbf{r}_{IC_{s,k}} \\ 1 \end{bmatrix} = \mathcal{I} \mathbf{T}_{\mathcal{S}_s} \begin{bmatrix} \mathcal{S}_s \mathbf{r}_{S_s C_{s,k}}^{\text{grid}} \\ 1 \end{bmatrix}.$$

Equivalently, in non-homogeneous form,

$$\mathcal{I} \mathbf{r}_{IC_{s,k}} = \mathcal{I} \mathbf{r}_{IS_s} + \mathbf{R}(\theta_s) \mathcal{S}_s \mathbf{r}_{S_s C_{s,k}}^{\text{grid}}. \quad (2.6)$$

From this point onward, world coordinates are the default. Thus $\mathbf{r}_{s,k}$ denotes the world-frame CoP $\mathcal{I} \mathbf{r}_{IC_{s,k}}$ unless a different frame is explicitly shown.

Each physical plate also defines a bounded spatial support. With side length L_s , its local domain is

$$\mathcal{S}_s \mathcal{B}_s = \left\{ \mathbf{r} \in \mathbb{R}^2 : \|\mathbf{r}\|_\infty \leq \frac{L_s}{2} \right\}.$$

The corresponding world-frame plate domain is

$$\mathcal{I} \mathcal{B}_s = \{ \mathcal{I} \mathbf{r}_{IS_s} + \mathbf{R}(\theta_s) \mathbf{r} : \mathbf{r} \in \mathcal{S}_s \mathcal{B}_s \}.$$

Hereafter, $\mathcal{B}_s$ denotes this world-frame plate domain; the frame-qualified form is used only when distinguishing it from the local plate domain.

The complete instrumented floor region is

$$\mathcal{B}^{\text{grid}} := \bigcup_{s \in S} \mathcal{B}_s,$$

with outer boundary $\partial \mathcal{B}^{\text{grid}}$.

In addition to the continuous plate geometry, each plate is associated with a discrete grid coordinate (i_s, j_s) . This allows nearby plates to be identified without repeated geometric distance calculations. For a nonnegative integer h_{grid} , the corresponding plate neighborhood is defined using the Chebyshev grid distance as

$$\mathcal{N}_{h_{\text{grid}}}(s) = \{ s' \in S : \max(|i_{s'} - i_s|, |j_{s'} - j_s|) \leq h_{\text{grid}} \}.$$

After filtering, thresholding, and transformation into the world frame, every active plate produces one initial force footprint. The set of active footprints at frame k is therefore

$$\mathcal{P}_k = \{ p(s, k) : s \in S, F_{s,k} \geq F_{\min} \}.$$

For $p = p(s, k)$, define its force, corner forces, and position by $F_{p,k} := F_{s,k}$, $F_{p,\ell,k} := F_{s\ell,k}^{\text{filt}}$, and $\mathbf{r}_{p,k} := \mathbf{r}_{s,k}$. Let $s(p)$ denote its source plate and $\mathcal{B}_p := \mathcal{B}_{s(p)}$ its admissible physical support. The footprint identifier inherits the source-sensor identifier, so the same p can be matched across frames in temporal terms; p is absent from $\mathcal{P}_k$ when its source plate is inactive.

Each resulting footprint contains the measured force $F_{p,k}$, the sub-plate CoP $\mathbf{r}_{p,k}$, the four corner forces $F_{p,\ell,k}$, and the physical support $\mathcal{B}_p$. This transformation converts the independent plates into a common spatial measurement field that can be used directly by the subsequent footprint allocation and tracking stages, as shown in fig. 2.3.

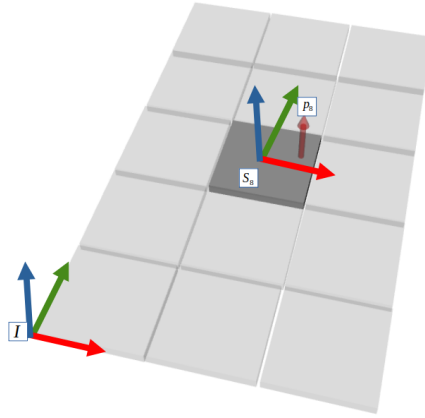


Figure 2.3: Force-floor spatial representation in the common world frame.

2.3 Global agent-tracking framework

The baseline tracker assumes no prior knowledge of how many entities are on the floor or what type of entities they are. At each sensor frame, active plates are represented by source footprints. The tracking lifecycle begins with footprints that have no persistent owner: they are grouped into blobs, accumulated across time, and promoted to agents after sufficient temporal evidence. Once an agent exists, its predicted state defines which current footprints are physically reachable. Unique interior claims are assigned directly, while edge plates and footprints claimed by more than one agent form ambiguity components that are resolved by the force-share and sub-grid position solvers. The accepted assignments update persistent agent states, and any remaining unassigned force is returned to the same blob-birth process. This ordering separates agent birth from agent maintenance. Birth requires only spatiotemporal consistency of unassigned force. Maintenance then exploits the agent state accumulated in previous frames to constrain ownership, motion, and force evolution.

2.3.1 Unassigned footprints and blob formation

Before any agent has been established, every active source footprint belongs to the unassigned set. After agents exist, the same set also receives source footprints with no claimant and the unassigned remainder of footprints that are only partially explained by existing agents. Let $u_{p,k} \in [0, 1]$ denote the unassigned fraction of source footprint p . The corresponding unassigned force is

$$F_{p,k}^u = u_{p,k} F_{p,k}, \quad \mathcal{P}_k^u = \{p \in \mathcal{P}_k : F_{p,k}^u > 0\}.$$

When no agent claims p , $u_{p,k} = 1$ and the complete footprint enters $\mathcal{P}_k^u$. For a partially shared source footprint, its unassigned position $\mathbf{r}_{p,k}^u$ is obtained from the sub-grid position solve; otherwise the measured CoP $\mathbf{r}_{p,k}$ is used.

The unassigned footprints are grouped spatially using DBSCAN [18] with neighborhood radius $r_c > 0$ and minimum sample count $n_{\text{DB}} = 1$. Thus every point is retained, and two unassigned footprints are neighbors when

$$\|\mathbf{r}_{p,k}^u - \mathbf{r}_{q,k}^u\|_2 \leq r_c.$$

Each resulting connected group defines a blob support $\mathcal{P}_{b,k}^u \subseteq \mathcal{P}_k^u$. Its total force, force-weighted centroid, and covariance are

$$\begin{aligned} F_{b,k} &= \sum_{p \in \mathcal{P}_{b,k}^u} F_{p,k}^u, \\ \boldsymbol{\mu}_{b,k} &= \sum_{p \in \mathcal{P}_{b,k}^u} w_{p,k} \mathbf{r}_{p,k}^u, \quad w_{p,k} = \frac{F_{p,k}^u}{F_{b,k}}, \\ \boldsymbol{\Sigma}_{b,k} &= \sum_{p \in \mathcal{P}_{b,k}^u} w_{p,k} (\mathbf{r}_{p,k}^u - \boldsymbol{\mu}_{b,k})(\mathbf{r}_{p,k}^u - \boldsymbol{\mu}_{b,k})^\top + \boldsymbol{\Sigma}_{\text{pos}}, \end{aligned}$$

where $\boldsymbol{\Sigma}_{\text{pos}}$ is the position measurement covariance obtained experimentally in section 3.2. The plate-support pattern and occupied footprint set are retained together with these force and position statistics, providing a compact description of each currently unassigned region.

2.3.2 Blob-track continuity and agent birth

An LMB birth layer [15] accumulates the blob sequence across frames. Each temporary track summarizes the recent position, velocity, force, spatial support,

and existence probability of one unassigned region. Association uses consistency in position, force, and occupied plate support, allowing short occlusions and temporary blob mergers to be bridged without requiring an agent identity.

A blob track becomes eligible to initialize an agent after it has remained sufficiently persistent and informative. Let n_b be its number of associated frames, $r_{b,k}^{\text{LMB}}$ its LMB existence probability at an associated frame k , and $\bar{F}_b$ its mean force over those frames. With $\bar{r}_b^{\text{LMB}} := n_b^{-1} \sum_{k=1}^{n_b} r_{b,k}^{\text{LMB}}$, the birth gate is

$$n_b \geq n_{\min}, \quad \bar{r}_b^{\text{LMB}} \geq r_{\min}^{\text{LMB}}, \quad \bar{F}_b \geq F_{\min}^{\text{birth}}$$

Here $n_{\min} \geq 1$, $r_{\min}^{\text{LMB}} \in (0, 1]$, and $F_{\min}^{\text{birth}} \geq F_{\min}$ are fixed birth hyperparameters. A confirmed track initializes the new agent position, force, uncertainty, and support from its current blob estimate. The temporary birth track is then terminated, and subsequent frames are handled by the persistent agent pipeline in sections 2.3.3, 2.3.4 and 2.3.8.

2.3.3 Persistent agent state and prediction

After birth, each agent $a \in \mathcal{A}_k$ maintains the filtered state

$$\mathbf{x}_{a,k} = \begin{bmatrix} r_{a,k}^x & r_{a,k}^y & v_{a,k}^x & v_{a,k}^y & F_{a,k} & \dot{F}_{a,k} \end{bmatrix}^T.$$

Its first two elements form the current world-frame agent position,

$$\mathbf{r}_{a,k} = [r_{a,k}^x \ r_{a,k}^y]^T.$$

Let $\mathbf{P}_{a,k}$ denote the covariance of $\mathbf{x}_{a,k}$. Before current-frame claims are constructed, this state is propagated with a constant-velocity and constant-force-rate model:

$$\mathbf{A}^A = \begin{bmatrix} 1 & 0 & \Delta t_s & 0 & 0 & 0 \\ 0 & 1 & 0 & \Delta t_s & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t_s \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H}^A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

$$\mathbf{x}_{a,k}^- = \mathbf{A}^A \mathbf{x}_{a,k-1}, \quad \mathbf{P}_{a,k}^- = \mathbf{A}^A \mathbf{P}_{a,k-1} (\mathbf{A}^A)^T + \mathbf{Q}^A,$$

where $\mathbf{Q}^A \succeq \mathbf{0}$ is the process-noise covariance and contains constant-acceleration blocks for planar motion and force rate. The predicted agent position $\mathbf{r}_{a,k}^-$ is the first two elements of $\mathbf{x}_{a,k}^-$; the superscript $-$ denotes a prior available before current-frame assignment. The accepted assignments update this predicted state later in section 2.3.8.

2.3.4 Claims and connected components

Because an agent's contribution to an aggregated footprint may originate anywhere within the plate area, claim eligibility uses the distance from the predicted agent position to the full plate support. A CoP-only distance could exclude a valid contribution elsewhere on the plate. For a point $\mathbf{r}$ and plate sensor s , define

$$d_{\text{plate}}(\mathbf{r}, s) = \min_{\mathbf{y} \in \mathcal{B}_s} \|\mathbf{r} - \mathbf{y}\|_2.$$

The initial claim mask is

$$C_{ap,k} = \mathbb{I}\left[d_{\text{plate}}(\mathbf{r}_{a,k}^-, s(p)) \leq \sigma_a^{\text{spread}}\right],$$

where $\sigma_a^{\text{spread}} > 0$ defines the maximum admissible reach of agent a from its predicted position.

Claims are additionally filtered for compatibility with the agent's physical extent. If agent a already has an unambiguous anchor footprint p_0 , another non-established claim on footprint p is rejected when the separation between the anchor and the candidate plate exceeds the admissible stride,

$$d_{\text{plate}}(\mathbf{r}_{p_0,k}, s(p)) > \sigma_a^{\text{stride}},$$

where $\sigma_a^{\text{stride}} > 0$ is the maximum admissible anchor-to-plate distance for the agent profile.

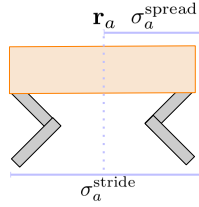


Figure 2.4: Spread and stride distances for a legged robot.

This prevents combinations of simultaneous claims that would require physically implausible support configurations. Accordingly, $C_{ap,k}$ represents only whether agent a is eligible to receive force from footprint p . The geometric construction is illustrated in fig. 2.5.

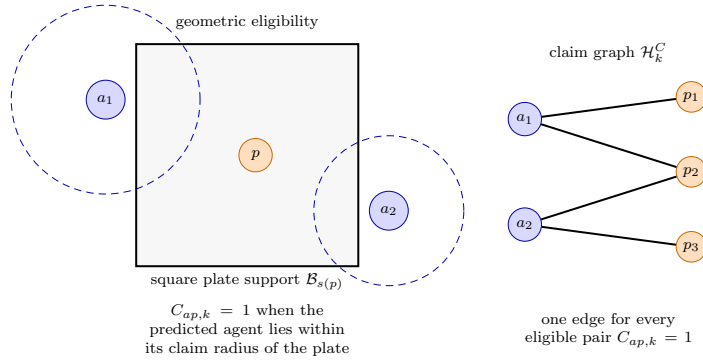


Figure 2.5: Claim construction. A predicted agent may claim a source footprint when it lies within its claim radius of the footprint's physical plate support. Eligible agent–footprint pairs form the bipartite claim graph; its connected components are solved independently.

An interior footprint with only one claimant is assigned directly. A footprint along the edge $\partial\mathcal{B}^{\text{grid}}$ remains in the optimization because part of its measured force may originate from an agent that is entering the instrumented floor from outside and is thus not yet tracked; this also provides the residual path used by force sustainment in section 2.3.7. Define the edge indicator

$$e_{p,k} = \begin{cases} 1, & \partial\mathcal{B}_{s(p)} \cap \partial\mathcal{B}^{\text{grid}} \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} \quad \mathbf{e}_k = [e_{p,k}]_{p \in \mathcal{P}_k}.$$

Thus $e_{p,k} = 1$ permits an explicit unassigned remainder for footprint p , whereas an interior footprint with $e_{p,k} = 0$ must be fully accounted for by its claiming agents. The set of footprints requiring joint optimization is therefore

$$\mathcal{P}_k^{\text{cont}} = \left\{ p \in \mathcal{P}_k : \sum_{a \in \mathcal{A}_k} C_{ap,k} > 0 \wedge \left(\sum_{a \in \mathcal{A}_k} C_{ap,k} > 1 \vee e_{p,k} = 1 \right) \right\}.$$

The outer condition excludes unclaimed footprints, which enter the unassigned set directly. The inner condition retains multi-agent claims and claimed boundary footprints for which an unassigned force share may be required. The corresponding bipartite claim graph is

$$\mathcal{A}_k^{\text{cont}} := \{ a \in \mathcal{A}_k : \exists p \in \mathcal{P}_k^{\text{cont}} \text{ with } C_{ap,k} = 1 \},$$

and

$$\mathcal{H}_k^C = (\mathcal{A}_k^{\text{cont}} \cup \mathcal{P}_k^{\text{cont}}, \{(a, p) : C_{ap,k} = 1, p \in \mathcal{P}_k^{\text{cont}}\}).$$

Every connected component is solved globally; distinct components share no decision variables or accounting constraints and can therefore be solved in parallel. Footprints with no claim enter the unassigned set in section 2.3.1; after the contested components are solved, any remaining residual share follows the same path.

2.3.5 Global fractional force assignment

After claim-graph decomposition, each connected component is solved as a convex quadratic program (QP) [26]. The QP determines the force fraction of each contested footprint assigned to each eligible agent; the positions at which these shares act are recovered by the position solver in section 2.3.6.

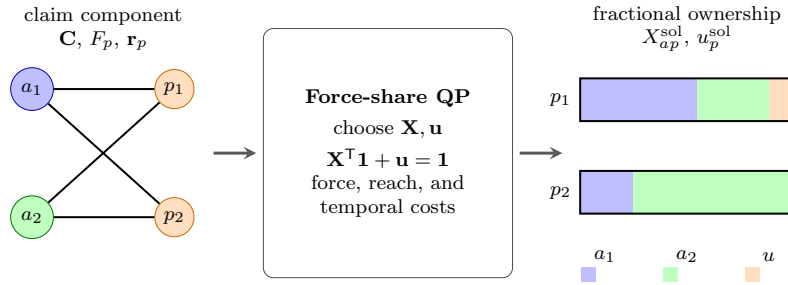


Figure 2.6: Force-share QP. An eligible agent–footprint claim component is optimized jointly, producing fractional ownership of each measured source footprint. The bar widths represent the solved shares; the source forces and measured CoPs remain fixed for the subsequent position solve.

For one connected component at sensor frame k , let

$$\mathcal{A} \subseteq \mathcal{A}_k, \quad \mathcal{P} \subseteq \mathcal{P}_k$$

denote its participating agents and contested source footprints, with

$$A = |\mathcal{A}|, \quad P = |\mathcal{P}|.$$

Only current-frame ownership shares are optimized; states from $k - 1$ and $k - 2$ are fixed historical anchors. The current-frame index is therefore omitted unless required.

Contested and fixed footprint contributions. For each agent $a \in \mathcal{A}$, let

$$\mathcal{P}_a^{\text{fix}} = \left\{ p \in \mathcal{P}_k : C_{ap,k} = 1, \sum_{d \in \mathcal{A}_k} C_{dp,k} = 1, e_{p,k} = 0 \right\}$$

be the set of current footprints assigned unambiguously to a .

Their forces $F_{p'}$ and world-frame CoPs $\mathbf{r}_{p'}$ are fixed inputs and are excluded from the contested set $\mathcal{P}$. Thus each agent's support consists of fixed support $\mathcal{P}_a^{\text{fix}}$ and contested support $\mathcal{P}$. The corresponding physical fixed force is $F_a^{\text{fix}} := \sum_{p' \in \mathcal{P}_a^{\text{fix}}} F_{p'}$. For every contested footprint $p \in \mathcal{P}$,

$$F_p > 0, \quad \mathbf{r}_p \in \mathbb{R}^2$$

are its measured vertical force and world-frame CoP.

Assignment target. Let $F_a^0 > 0$ be the learned neutral force of agent a , initialized at birth and updated in section 2.3.8, and let $\mathcal{D}_a^{\text{carry}}$ contain agents or objects currently carried by a as defined in section 2.4.4. The assignment target is

$$N_a = F_a^0 + \sum_{d \in \mathcal{D}_a^{\text{carry}}} F_d^0. \quad (2.7)$$

If $\mathcal{D}_a^{\text{carry}} = \emptyset$, then $N_a = F_a^0$. Treating the target as a soft cost accommodates gait, impacts, load transfer, and other transient effects.

Fractional ownership variables. For each $(a, p) \in \mathcal{A} \times \mathcal{P}$, define

$$X_{ap} \in [0, 1]$$

as the fraction of footprint p assigned to agent a , giving

$$F_{ap} = X_{ap} F_p.$$

Thus $X_{ap} = 0$ assigns none of the footprint, $X_{ap} = 1$ assigns all of it, and $0 < X_{ap} < 1$ represents sharing. Collecting all shares,

$$\mathbf{X} = [X_{ap}] \in \mathbb{R}^{A \times P}.$$

Each footprint may retain an unassigned fraction

$$u_p \in [0, 1],$$

collected as

$$\mathbf{u} = [u_p] \in \mathbb{R}^P.$$

This residual force remains represented as an unassigned child footprint and may subsequently enter the residual-footprint clustering and agent-birth pipeline (section 2.3.1).

Claim and residual eligibility. The fixed claim matrix is

$$\mathbf{C} = [C_{ap}] \in \{0, 1\}^{A \times P},$$

with

$$C_{ap} = \begin{cases} 1, & \text{if agent } a \text{ may claim footprint } p, \\ 0, & \text{otherwise.} \end{cases}$$

The bound

$$0 \leq X_{ap} \leq C_{ap}$$

therefore excludes invalid agent–footprint assignments.

Similarly,

$$\mathbf{e} = [e_p] \in \{0, 1\}^P$$

defines whether footprint p comes from a plate at the edge of the sensed region and thus may retain an explicit residual, with

$$0 \leq u_p \leq e_p.$$

Only plates on the boundary of the sensed region may leave footprint shares unassigned, representing an agent entering the instrumented floor.

Exact force accounting requires

$$\sum_{a \in \mathcal{A}} X_{ap} + u_p = 1, \quad \forall p \in \mathcal{P}.$$

Hence the complete ownership constraints are

$$\mathbf{0} \preceq \mathbf{X} \preceq \mathbf{C}, \quad \mathbf{0} \preceq \mathbf{u} \preceq \mathbf{e}, \quad \mathbf{X}^\top \mathbf{1}_A + \mathbf{u} = \mathbf{1}_P. \quad (2.8)$$

All vector and matrix inequalities in this QP are componentwise.

Every footprint must have at least one legal destination:

$$\mathbf{C}^\top \mathbf{1}_A + \mathbf{e} \succeq \mathbf{1}_P. \quad (2.9)$$

Decision variables. The complete decision block is

$$\mathbf{X} \in \mathbb{R}^{A \times P}, \quad \mathbf{u} \in \mathbb{R}^P, \quad \mathbf{s}^+, \mathbf{s}^-, \boldsymbol{\rho}^N, \boldsymbol{\rho}^E \in \mathbb{R}^A.$$

The non-negative vectors $\mathbf{s}^+$ and $\mathbf{s}^-$ relax the upper and lower force envelopes. $\boldsymbol{\rho}^N = [\rho_a^N]$ is the normalized assignment-target mismatch, and $\boldsymbol{\rho}^E = [\rho_a^E]$ is the one-sided excess-force residual.

Component normalization. Force, position, and first moment are normalized independently. Define

$$\sigma_F = \max\left(1 \text{ N}, \max_{a \in \mathcal{A}} N_a\right),$$

with normalized forces

$$\hat{F}_p = \frac{F_p}{\sigma_F}, \quad \hat{N}_a = \frac{N_a}{\sigma_F}.$$

The component origin is

$$\mathbf{r}_0 = \frac{1}{P} \sum_{p \in \mathcal{P}} \mathbf{r}_p,$$

and the position scale is

$$\sigma_r = \max\left(1 \text{ m}, \max_{p \in \mathcal{P}} \|\mathbf{r}_p - \mathbf{r}_0\|_2\right).$$

The normalized footprint position is

$$\hat{\mathbf{r}}_p = \frac{\mathbf{r}_p - \mathbf{r}_0}{\sigma_r},$$

and its normalized first moment is

$$\hat{\mathbf{m}}_p = \hat{F}_p \hat{\mathbf{r}}_p \in \mathbb{R}^2.$$

All historical forces and positions are re-expressed using the current σ_F , $\mathbf{r}_0$, and σ_r so that temporal residuals use a common scale and frame.

Fixed force and first-moment contribution. For every $p' \in \mathcal{P}_a^{\text{fix}}$,

$$\hat{F}_{p'} = \frac{F_{p'}}{\sigma_F}, \quad \hat{\mathbf{r}}_{p'} = \frac{\mathbf{r}_{p'} - \mathbf{r}_0}{\sigma_r}, \quad \hat{\mathbf{m}}_{p'} = \hat{F}_{p'} \hat{\mathbf{r}}_{p'}.$$

The fixed normalized force and first moment of agent a are

$$\hat{F}_a^{\text{fix}} = \sum_{p' \in \mathcal{P}_a^{\text{fix}}} \hat{F}_{p'},$$

and

$$\hat{\mathbf{m}}_a^{\text{fix}} = \sum_{p' \in \mathcal{P}_a^{\text{fix}}} \hat{F}_{p'} \hat{\mathbf{r}}_{p'}.$$

Both are zero when $\mathcal{P}_a^{\text{fix}} = \emptyset$. They ensure that contested-force allocation is evaluated together with support already owned by the agent.

Stack the fixed forces as

$$\hat{\mathbf{f}}_A^{\text{fix}} = [\hat{F}_a^{\text{fix}}]_{a \in \mathcal{A}} \in \mathbb{R}^A,$$

and the fixed first moments as

$$\hat{\mathbf{M}}_A^{\text{fix}} \in \mathbb{R}^{A \times 2}, \quad [\hat{\mathbf{M}}_A^{\text{fix}}]_{a,:} = (\hat{\mathbf{m}}_a^{\text{fix}})^\top.$$

Similarly,

$$\hat{\mathbf{f}}_P = [\hat{F}_p]_{p \in \mathcal{P}} \in \mathbb{R}^P$$

and

$$\hat{\mathbf{M}}_P \in \mathbb{R}^{P \times 2}, \quad [\hat{\mathbf{M}}_P]_{p,:} = \hat{\mathbf{m}}_p^\top.$$

Current agent force and first moment. For candidate ownership matrix $\mathbf{X}$,

$$\hat{\mathbf{f}}_A = \mathbf{X} \hat{\mathbf{f}}_P + \hat{\mathbf{f}}_A^{\text{fix}}, \quad (2.10)$$

or

$$\hat{F}_a = \hat{F}_a^{\text{fix}} + \sum_{p \in \mathcal{P}} X_{ap} \hat{F}_p.$$

Likewise,

$$\hat{\mathbf{M}}_A = \mathbf{X} \hat{\mathbf{M}}_P + \hat{\mathbf{M}}_A^{\text{fix}}, \quad (2.11)$$

or

$$\hat{\mathbf{m}}_a = \hat{\mathbf{m}}_a^{\text{fix}} + \sum_{p \in \mathcal{P}} X_{ap} \hat{F}_p \hat{\mathbf{r}}_p.$$

Both are affine in $\mathbf{X}$.

Agent centroid and affine approximation. For $\hat{F}_a > 0$, the exact normalized force-weighted centroid is

$$\hat{\mathbf{r}}_a^{\text{exact}} = \frac{\hat{\mathbf{m}}_a}{\hat{F}_a}.$$

Because both numerator and denominator depend on $\mathbf{X}$, this ratio is nonlinear. It is therefore linearized around the accepted state at $k-1$.

Let

$$\hat{F}_{a,k-1}, \quad \hat{\mathbf{r}}_{a,k-1}, \quad \hat{\mathbf{m}}_{a,k-1}$$

be the previous force, centroid, and first moment expressed using the current normalization, with

$$\hat{\mathbf{m}}_{a,k-1} = \hat{F}_{a,k-1} \hat{\mathbf{r}}_{a,k-1}.$$

Define

$$\iota_a = \begin{cases} 1/\hat{F}_{a,k-1}, & \hat{F}_{a,k-1} > \varepsilon_{\hat{F}}, \\ 0, & \hat{F}_{a,k-1} \leq \varepsilon_{\hat{F}}, \end{cases}$$

where $0 < \varepsilon_{\hat{F}} \ll 1$ is a dimensionless numerical tolerance that prevents division by near-zero historical force. The first-order approximation is

$$\begin{aligned} \hat{\mathbf{r}}_a^{\text{lin}} &= \hat{\mathbf{r}}_{a,k-1} + \iota_a (\hat{\mathbf{m}}_a - \hat{\mathbf{m}}_{a,k-1}) \\ &\quad - \iota_a \hat{\mathbf{r}}_{a,k-1} (\hat{F}_a - \hat{F}_{a,k-1}). \end{aligned} \quad (2.12)$$

Since $\hat{\mathbf{m}}_a$ and $\hat{F}_a$ are affine in $\mathbf{X}$, so is $\hat{\mathbf{r}}_a^{\text{lin}}$. If $\hat{F}_{a,k-1} \leq \varepsilon_{\hat{F}}$, then $\iota_a = 0$ and the approximation reduces to the previous position.

Temporal anchors and current-frame curvature. For scalar or vector state q ,

$$\Delta^2 q_k = q_k - 2q_{k-1} + q_{k-2}.$$

Only q_k depends on the current optimization.

For position,

$$\boldsymbol{\rho}_a^R = \hat{\mathbf{r}}_a^{\text{lin}} - 2\hat{\mathbf{r}}_{a,k-1} + \hat{\mathbf{r}}_{a,k-2} \in \mathbb{R}^2,$$

which is approximately zero under constant-velocity motion.

For total assigned force,

$$\rho_a^F = \hat{F}_a - 2\hat{F}_{a,k-1} + \hat{F}_{a,k-2}.$$

The current normalized force allocated from footprint p to agent a is

$$\hat{F}_{ap} = X_{ap} \hat{F}_p.$$

Let $\hat{F}_{ap,k-1}$ and $\hat{F}_{ap,k-2}$ be the corresponding accepted historical allocations, expressed using the current σ_F and defined as zero when the pair did not exist. Then

$$\rho_{ap}^{PF} = X_{ap} \hat{F}_p - 2\hat{F}_{ap,k-1} + \hat{F}_{ap,k-2}.$$

This penalizes unnecessary redistribution of a footprint between agents across frames.

Spatial spread cost. For each pair (a, p) , define

$$D_{ap} = \lambda_a^{\text{spread}} \|\hat{\mathbf{r}}_{a,k-1} - \hat{\mathbf{r}}_p\|_2^2,$$

where

$$\lambda_a^{\text{spread}} > 0$$

is a fixed dimensionless agent-specific spread coefficient. Larger values penalize spatially distant ownership more strongly. This allows agents with known, elongated form factors to claim footprints that are further away, while favoring more compact agents to claim nearby footprints.

Collecting the costs,

$$\mathbf{D} = [D_{ap}] \in \mathbb{R}^{A \times P}.$$

The total spread cost is

$$\langle \mathbf{X}, \mathbf{D} \rangle_F = \sum_{a \in \mathcal{A}} \sum_{p \in \mathcal{P}} X_{ap} D_{ap}.$$

Claiming determines admissibility through $\mathbf{C}$; the spread term ranks admissible assignments.

Assignment-target mismatch. Define

$$\hat{N}_a \rho_a^N = \hat{F}_a - \hat{N}_a. \quad (2.13)$$

For $\hat{N}_a > 0$,

$$\rho_a^N = \frac{\hat{F}_a - \hat{N}_a}{\hat{N}_a}.$$

Thus $\rho_a^N = 0$ matches the target, while positive and negative values represent excess and insufficient force.

Dynamic force envelope and slack variables. Each agent has a current force envelope

$$[F_a^-, F_a^+], \quad 0 \leq F_a^- \leq F_a^+,$$

with normalized limits

$$\hat{F}_a^- = \frac{F_a^-}{\sigma_F}, \quad \hat{F}_a^+ = \frac{F_a^+}{\sigma_F}.$$

Introduce

$$s_a^- \geq 0, \quad s_a^+ \geq 0,$$

and impose

$$\hat{F}_a^- - s_a^- \leq \hat{F}_a \leq \hat{F}_a^+ + s_a^+. \quad (2.14)$$

The slacks preserve feasibility when measured force legitimately exceeds the nominal envelope, which can occur due to noise introduced by the sensors or environmental variables.

Sustained excess residual. Define

$$\rho_a^E \geq 0, \quad \rho_a^E \geq \hat{F}_a - \hat{N}_a. \quad (2.15)$$

Whenever $\lambda_E \gamma_a^2 > 0$, optimality gives

$$\rho_a^E = \max(\hat{F}_a - \hat{N}_a, 0).$$

If $\gamma_a = 0$, this auxiliary variable does not affect the ownership solution and its smallest feasible value is selected after solving.

Let

$$\gamma_a \geq 0$$

be an externally maintained dimensionless multiplier representing accumulated evidence of persistent unexplained excess force, and collect

$$\boldsymbol{\gamma} = [\gamma_a] \in \mathbb{R}^A.$$

The residual deliberately distinguishes a high force in one frame from a high force sustained across frames. Transient excess can arise naturally during dynamic motion, such as gait, impacts, or load transfer, and is therefore accommodated by the soft force envelope and the instantaneous residual. In contrast, a persistently positive excess residual increases γ_a and is treated as evidence that the existing agents do not fully explain the measured load. The solver then favors representing that force as an explicit residual, which can enter the agent-birth pipeline as a potential newly introduced agent. This is discussed further in section 2.3.7.

Individual objective terms. The assignment-target cost is

$$\mathcal{L}_N = \sum_{a \in \mathcal{A}} (\rho_a^N)^2 = \|\boldsymbol{\rho}^N\|_2^2.$$

The spatial spread cost is

$$\mathcal{L}_S = \langle \mathbf{X}, \mathbf{D} \rangle_F.$$

The unassigned-share cost is

$$\mathcal{L}_U = \sum_{p \in \mathcal{P}} u_p = \mathbf{1}_P^\top \mathbf{u},$$

which is kept linear because a quadratic penalty $\sum_{p \in \mathcal{P}} u_p^2$ would favor distributing a given residual across multiple footprints.

The position-curvature cost is

$$\mathcal{L}_R = \sum_{a \in \mathcal{A}} \|\boldsymbol{\rho}_a^R\|_2^2.$$

The total-force curvature cost is

$$\mathcal{L}_F = \sum_{a \in \mathcal{A}} (\rho_a^F)^2.$$

The pair-force curvature cost is

$$\mathcal{L}_{PF} = \sum_{a \in \mathcal{A}} \sum_{p \in \mathcal{P}} (\rho_{ap}^{PF})^2.$$

The soft-envelope cost is

$$\mathcal{L}_B = \|\mathbf{s}^+\|_2^2 + \|\mathbf{s}^-\|_2^2.$$

The sustained-excess cost is

$$\mathcal{L}_E = \|\text{diag}(\boldsymbol{\gamma}) \boldsymbol{\rho}^E\|_2^2 = \sum_{a \in \mathcal{A}} (\gamma_a \rho_a^E)^2.$$

These terms respectively favor the expected agent load, compact ownership, low residual force, smooth support motion, smooth total force, stable pairwise ownership, limited envelope violations, and low persistent unexplained excess.

Complete quadratic program. Let

$$\lambda_N, \lambda_S, \lambda_U, \lambda_R, \lambda_F, \lambda_{PF}, \lambda_B, \lambda_E \geq 0$$

be fixed dimensionless objective weights. The complete objective is

$$\begin{aligned} \mathcal{L}_{\text{assign}} = & \lambda_N \mathcal{L}_N + \lambda_S \mathcal{L}_S + \lambda_U \mathcal{L}_U + \lambda_R \mathcal{L}_R \\ & + \lambda_F \mathcal{L}_F + \lambda_{PF} \mathcal{L}_{PF} + \lambda_B \mathcal{L}_B + \lambda_E \mathcal{L}_E. \end{aligned} \quad (2.16)$$

The solution is

$$(\mathbf{X}^{\text{sol}}, \mathbf{u}^{\text{sol}}, \mathbf{s}^{+, \text{sol}}, \mathbf{s}^{-, \text{sol}}, \boldsymbol{\rho}^{N, \text{sol}}, \boldsymbol{\rho}^{E, \text{sol}}) \in \arg \min_{\mathbf{X}, \mathbf{u}, \mathbf{s}^+, \mathbf{s}^-, \boldsymbol{\rho}^N, \boldsymbol{\rho}^E} \mathcal{L}_{\text{assign}}$$

subject to the ownership constraints in eq. (2.8), the affine force and moment definitions in eqs. (2.10) and (2.11), the centroid approximation in eq. (2.12), and the target, envelope, and excess relations in eqs. (2.13) to (2.15). The coverage condition in eq. (2.9) is checked before solving.

$\mathbf{X}^{\text{sol}}$ gives the agent ownership fractions and $\mathbf{u}^{\text{sol}}$ the explicit unassigned fractions; the remaining variables are auxiliary optimization quantities.

Convexity. All measurements, eligibility masks, historical states, fixed contributions, force targets, bounds, normalizers, spread coefficients, sustained-excess multipliers, and weights are fixed before solving. In particular,

$$\begin{aligned} &F_p, \mathbf{r}_p, \mathbf{C}, \mathbf{e}, N_a, \mathcal{P}_a^{\text{fix}}, \hat{F}_a^{\text{fix}}, \hat{\mathbf{m}}_a^{\text{fix}}, \\ &\hat{\mathbf{r}}_{a,k-1}, \hat{\mathbf{r}}_{a,k-2}, \hat{F}_{a,k-1}, \hat{F}_{a,k-2}, \hat{F}_{ap,k-1}, \hat{F}_{ap,k-2}, \\ &\lambda_a^{\text{spread}}, F_a^-, F_a^+, \gamma_a \end{aligned}$$

are parameters.

Consequently, $\hat{F}_a$, $\hat{\mathbf{m}}_a$, $\hat{\mathbf{r}}_a^{\text{lin}}$, ρ_a^R , ρ_a^F , and ρ_{ap}^{PF} are affine in the decision variables. Their squared norms are convex quadratic, while $\langle \mathbf{X}, \mathbf{D} \rangle_F$ and $\mathbf{1}_p^\top \mathbf{u}$ are linear. All constraints are affine, so nonnegative weights yield a convex QP. The first-order centroid approximation is what preserves this structure.

2.3.6 Sub-grid position QP

The force-share QP determines the force assigned to each participant in a contested source footprint but not the positions at which those shares act. A second QP is therefore solved independently for every shared source footprint to recover one planar contact position per non-zero share while preserving the measured total force and first moment.

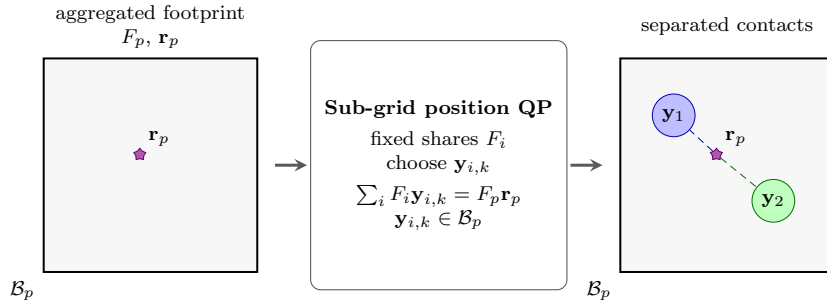


Figure 2.7: Sub-grid position QP. The fixed force shares of one aggregated footprint are assigned separate contact positions inside the same plate. Their force-weighted centroid (star) remains exactly equal to the measured CoP, so total force and first moment are conserved.

Participants created by the force-share solution. The preceding force-share solution

$$X_{ap}^{\text{sol}}, \quad u_p^{\text{sol}}$$

is fixed before this QP. Shares whose force is at most F_ϵ are set to zero to filter out noise introduced by the force share solver. The remaining shares of footprint p are renormalized to sum to one. For an agent share,

$$F_{ap} = X_{ap}^{\text{sol}} F_p,$$

and for an unassigned remainder,

$$F_{up} = u_p^{\text{sol}} F_p.$$

Every retained positive share is a participant:

$$\mathcal{I}_p = \{ap : X_{ap}^{\text{sol}} > 0\} \cup \{up : u_p^{\text{sol}} > 0\}.$$

The composite index i denotes either an agent share ap or the unassigned share up , with

$$F_i = \begin{cases} F_{ap}, & i = ap, \\ F_{up}, & i = up. \end{cases}$$

All F_i are fixed.

Since

$$\sum_a X_{ap}^{\text{sol}} + u_p^{\text{sol}} = 1,$$

the participant forces satisfy

$$\sum_{i \in \mathcal{I}_p} F_i = F_p.$$

Contact-position decision variables. For each $i \in \mathcal{I}_p$, define

$$\mathbf{y}_{i,k} = \begin{bmatrix} y_{i,k}^x \\ y_{i,k}^y \end{bmatrix} \in \mathbb{R}^2.$$

Each position must remain inside the source plate:

$$\mathbf{y}_{i,k} \in \mathcal{B}_p, \quad \forall i \in \mathcal{I}_p.$$

First-moment conservation. The measured first moment is

$$F_p \mathbf{r}_p.$$

The reconstructed participant moment is

$$\sum_{i \in \mathcal{I}_p} F_i \mathbf{y}_{i,k}.$$

Exact preservation requires

$$\sum_{i \in \mathcal{I}_p} F_i \mathbf{y}_{i,k} = F_p \mathbf{r}_p.$$

Together with force conservation,

$$\frac{\sum_{i \in \mathcal{I}_p} F_i \mathbf{y}_{i,k}}{\sum_{i \in \mathcal{I}_p} F_i} = \mathbf{r}_p.$$

Thus the reconstructed force-weighted centroid exactly equals the measured CoP.

If only one participant exists,

$$\mathbf{y}_{i,k} = \mathbf{r}_p.$$

Directional and referenced participants. Partition

$$\mathcal{I}_p = \mathcal{I}_p^{\text{dir}} \dot{\cup} \mathcal{I}_p^{\text{ref}},$$

where directional participants use geometric inference and referenced participants have trusted position references obtained from robot model contacts (section 2.4.1), external sensors, or scene-specific assumptions such as static object positions. For every $i \in \mathcal{I}_p^{\text{ref}}$,

$$\mathbf{y}_i^{\text{ref}} \in \mathcal{B}_p$$

is its trusted target. It remains a soft target so that exact first-moment conservation can still be satisfied in the presence of measurement noise.

For every $i \in \mathcal{I}_p^{\text{dir}}$, no direct contact-position reference is available. A directional proxy reduces the underdetermined solution space while retaining a geometrically plausible decomposition.

Directional anchors. For an agent participant $i = ap$, define

$$\mathbf{a}_i = \mathbf{r}_{a,k}^-$$

as the predicted current position of agent a before current footprint assignment. For the unassigned participant, the anchor is chosen as the closest point on the outer boundary $\partial\mathcal{B}^{\text{grid}}$ defined in section 2.2.4. The residual anchor is

$$\mathbf{a}_{up} = \text{lexmin} \left(\arg \min_{\mathbf{y} \in \partial\mathcal{B}^{\text{grid}}} \|\mathbf{y} - \mathbf{r}_p\|_2 \right),$$

where lexmin resolves equidistant solutions deterministically.

For every directional participant, let $0 < \varepsilon_r \ll 1$ be a position tolerance and define

$$\mathbf{v}_i = \begin{cases} \frac{\mathbf{a}_i - \mathbf{r}_p}{\|\mathbf{a}_i - \mathbf{r}_p\|_2}, & \|\mathbf{a}_i - \mathbf{r}_p\|_2 > \varepsilon_r, \\ \mathbf{0}, & \text{otherwise,} \end{cases} \quad i \in \mathcal{I}_p^{\text{dir}},$$

These fixed vectors encode preferred directions from the parent CoP; the zero vector represents a coincident anchor and receives the parent CoP as its target.

Directional partition of the footprint support. For each directional participant with $\mathbf{v}_i \neq \mathbf{0}$,

$$\mathcal{R}_i = \mathcal{B}_p \cap \bigcap_{\substack{j \in \mathcal{I}_p^{\text{dir}}, \mathbf{v}_j \neq \mathbf{0} \\ j \neq i}} \{ \mathbf{y} \in \mathbb{R}^2 : (\mathbf{y} - \mathbf{r}_p)^\top (\mathbf{v}_i - \mathbf{v}_j) \geq 0 \}.$$

The pairwise condition

$$(\mathbf{y} - \mathbf{r}_p)^\top (\mathbf{v}_i - \mathbf{v}_j) \geq 0$$

selects points whose displacement from $\mathbf{r}_p$ aligns at least as strongly with $\mathbf{v}_i$ as with $\mathbf{v}_j$. For distinct directions, the resulting regions form a directional Voronoi-like partition about the common origin $\mathbf{r}_p$. Coincident directions produce the same region and target; temporal and diagonal costs then provide the remaining separation information.

The geometric centers of these regions generate soft guidance targets to limit the solution spaces.

For directional participants, define

$$\mathbf{c}_i = \begin{cases} \text{centroid}(\mathcal{R}_i), & \mathbf{v}_i \neq \mathbf{0}, \\ \mathbf{r}_p, & \mathbf{v}_i = \mathbf{0}, \end{cases} \quad i \in \mathcal{I}_p^{\text{dir}}.$$

An example with three distinct anchors is shown in fig. 2.8.

Temporal position anchors. Let

$$\mathbf{y}_{i,k-1}, \quad \mathbf{y}_{i,k-2}$$

be the accepted positions of the same participant in the two preceding frames. The position-curvature residual is

$$\rho_{i,k}^Y = \mathbf{y}_{i,k} - 2\mathbf{y}_{i,k-1} + \mathbf{y}_{i,k-2}.$$

When either of the two previous positions is unavailable, it is substituted by the positional target $\mathbf{c}_i$ or $\mathbf{y}_i^{\text{ref}}$. This ensures that the curvature term is always defined.

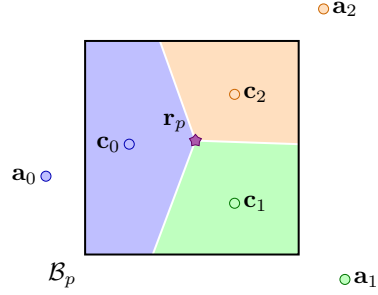


Figure 2.8: Directional regions and participant anchors. The agent anchors $\mathbf{a}_i$ generate the directions $\mathbf{v}_i$ from the measured CoP $\mathbf{r}_p$, producing the clipped directional cells $\mathcal{R}_i$ with centroid $\mathbf{c}_i$.

Information remaining in the four load-cell measurements. Let

$$F_{p,1}, \quad F_{p,2}, \quad F_{p,3}, \quad F_{p,4}$$

be the four calibrated load-cell forces. Their total is

$$F_p = F_{p,1} + F_{p,2} + F_{p,3} + F_{p,4}.$$

Total force and the two CoP coordinates account for three independent combinations, leaving one independent spatial loading mode. Define the checkerboard diagonal combination

$$\chi_p = F_{p,1} - F_{p,2} + F_{p,3} - F_{p,4}. \quad (2.17)$$

Opposite corners have equal sign, allowing χ_p to distinguish load arrangements with identical total force and CoP.

Normalized plate coordinates. Let

$$\boldsymbol{\xi}_p(\mathbf{y}) = \begin{bmatrix} \xi_p(\mathbf{y}) \\ v_p(\mathbf{y}) \end{bmatrix}$$

map $\mathbf{y} \in \mathcal{B}_p$ to normalized plate coordinates

$$\xi_p(\mathbf{y}), v_p(\mathbf{y}) \in [0, 1].$$

Equivalently,

$$\boldsymbol{\xi}_p(\mathbf{y}) = \mathbf{A}_p^\xi \mathbf{y} + \mathbf{b}_p^\xi,$$

where

$$\mathbf{A}_p^\xi \in \mathbb{R}^{2 \times 2}, \quad \mathbf{b}_p^\xi \in \mathbb{R}^2$$

are fixed by the plate pose and dimensions.

Define

$$h(\xi, v) = (1 - 2\xi)(1 - 2v).$$

At the unit-square corners,

$$h(0, 0) = +1, \quad h(1, 0) = -1, \quad h(1, 1) = +1, \quad h(0, 1) = -1,$$

matching the alternating signs in eq. (2.17).

For world-frame position $\mathbf{y}$,

$$h_p(\mathbf{y}) := h(\xi_p(\mathbf{y}), v_p(\mathbf{y})).$$

The exact predicted diagonal mode is

$$\chi_p^{\text{pred}} = \sum_{i \in \mathcal{I}_p} F_i h_p(\mathbf{y}_{i,k}).$$

It is nonlinear because $h(\xi, v)$ contains the product ξv .

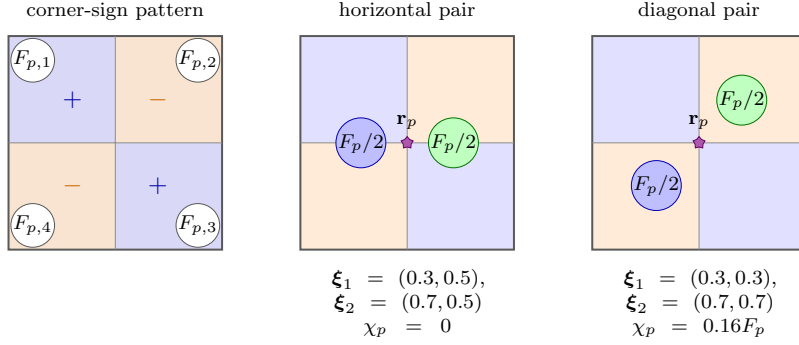


Figure 2.9: Two equal-force contact arrangements with the same measured CoP. The horizontal pair lies on the zero-response line of the diagonal mode, whereas the diagonal pair produces a nonzero χ_p . Thus the four load-cell readings retain a spatial distinction that total force and CoP alone do not provide..

Linearization of the diagonal response. Linearize about the positional target

$$\bar{\mathbf{y}}_i = \begin{cases} \mathbf{c}_i, & i \in \mathcal{I}_p^{\text{dir}}, \\ \mathbf{y}_i^{\text{ref}}, & i \in \mathcal{I}_p^{\text{ref}}. \end{cases}$$

These references are fixed during the solve.

Define

$$h_i := h_p(\bar{\mathbf{y}}_i)$$

and

$$\mathbf{g}_i^x := \nabla_{\mathbf{y}} h_p(\mathbf{y})|_{\mathbf{y}=\bar{\mathbf{y}}_i} \in \mathbb{R}^2.$$

The first-order approximation is

$$h_p(\mathbf{y}_{i,k}) \approx h_i + (\mathbf{g}_i^x)^\top (\mathbf{y}_{i,k} - \bar{\mathbf{y}}_i),$$

or equivalently

$$h_p(\mathbf{y}_{i,k}) \approx (\mathbf{g}_i^x)^\top \mathbf{y}_{i,k} + h_i - (\mathbf{g}_i^x)^\top \bar{\mathbf{y}}_i.$$

Substitution gives

$$\begin{aligned} \chi_p^{\text{lin}} &= \sum_{i \in \mathcal{I}_p} F_i (\mathbf{g}_i^x)^\top \mathbf{y}_{i,k} \\ &\quad + \sum_{i \in \mathcal{I}_p} F_i [h_i - (\mathbf{g}_i^x)^\top \bar{\mathbf{y}}_i]. \end{aligned} \quad (2.18)$$

Define

$$\kappa_\chi = \sum_{i \in \mathcal{I}_p} F_i [h_i - (\mathbf{g}_i^x)^\top \bar{\mathbf{y}}_i].$$

Then

$$\chi_p^{\text{lin}} = \sum_{i \in \mathcal{I}_p} F_i(\mathbf{g}_i^\chi)^\top \mathbf{y}_{i,k} + \kappa_\chi.$$

Matching the measured diagonal mode therefore gives

$$\sum_{i \in \mathcal{I}_p} F_i(\mathbf{g}_i^\chi)^\top \mathbf{y}_{i,k} - (\chi_p - \kappa_\chi) \approx 0.$$

Normalization of the diagonal residual. Define

$$\sigma_\chi = \max(F_p, 1 \text{ N}).$$

The normalized residual is

$$\rho_p^\chi = \sum_{i \in \mathcal{I}_p} \frac{F_i}{\sigma_\chi} (\mathbf{g}_i^\chi)^\top \mathbf{y}_{i,k} - \frac{\chi_p - \kappa_\chi}{\sigma_\chi}.$$

It is affine because F_i , σ_χ , $\mathbf{g}_i^\chi$, χ_p , and κ_χ are fixed. The equality $\rho_p^\chi = 0$ reproduces the measured diagonal mode under the first-order approximation.

Individual objective terms. The directional target cost is

$$\mathcal{L}_{\text{dir}} = \sum_{i \in \mathcal{I}_p^{\text{dir}}} \|\mathbf{y}_{i,k} - \mathbf{c}_i\|_2^2.$$

The referenced-contact cost is

$$\mathcal{L}_{\text{ref}} = \sum_{i \in \mathcal{I}_p^{\text{ref}}} \|\mathbf{y}_{i,k} - \mathbf{y}_i^{\text{ref}}\|_2^2.$$

The temporal smoothness cost is

$$\mathcal{L}_{\text{hist}} = \sum_{i \in \mathcal{I}_p} \|\boldsymbol{\rho}_{i,k}^Y\|_2^2.$$

The diagonal load-cell cost is

$$\mathcal{L}_{\text{diag}} = (\rho_p^\chi)^2.$$

These terms respectively favor directional separation, agreement with referenced contacts, temporally smooth contact motion, and consistency with the independent diagonal loading information retained by the four load-cell measurements.

Complete quadratic program. Let

$$\lambda_{\text{dir}}, \lambda_{\text{ref}}, \lambda_{\text{hist}}, \lambda_{\text{diag}} \geq 0$$

be fixed nonnegative objective weights. The three position weights have units m^{-2} and λ_{diag} is dimensionless, making the complete objective dimensionless. The objective for source footprint p is

$$\begin{aligned} \mathcal{L}_{\text{subgrid},p} &= \lambda_{\text{dir}} \mathcal{L}_{\text{dir}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} \\ &\quad + \lambda_{\text{hist}} \mathcal{L}_{\text{hist}} + \lambda_{\text{diag}} \mathcal{L}_{\text{diag}}. \end{aligned}$$

The solution is

$$\{\mathbf{y}_{i,k}^{\text{sol}}\}_{i \in \mathcal{I}_p} \in \arg \min_{\{\mathbf{y}_{i,k}\}_{i \in \mathcal{I}_p}} \mathcal{L}_{\text{subgrid},p}$$

subject to

$$\sum_{i \in \mathcal{I}_p} F_i \mathbf{y}_{i,k} = F_p \mathbf{r}_p, \quad \mathbf{y}_{i,k} \in \mathcal{B}_p \quad \forall i \in \mathcal{I}_p.$$

The first-moment equality and plate-support constraints enforce physical equivalence to the measured footprint. The decision variables are the resolved participant positions; all forces, targets, references, historical positions, and diagonal-measurement quantities are fixed before solving.

Convexity. At this stage,

$$F_p, \mathbf{r}_p, F_i, \mathcal{B}_p, \mathbf{a}_i, \mathbf{v}_i, \mathcal{R}_i, \mathbf{c}_i, \mathbf{y}_i^{\text{ref}}, \\ \mathbf{y}_{i,k-1}, \mathbf{y}_{i,k-2}, \chi_p, \sigma_\chi, \bar{\mathbf{y}}_i, h_i, \mathbf{g}_i^\chi, \kappa_\chi$$

are fixed parameters. The only decision variables are

$$\mathbf{y}_{i,k}, \quad i \in \mathcal{I}_p.$$

The first-moment constraint is affine, and membership in $\mathcal{B}_p$ is represented by the four linear half-spaces of the rotated square. The directional, referenced-contact, temporal, and linearized diagonal residuals are affine in the current positions, so their squared norms are convex quadratic. Although the exact diagonal function $h_p(\mathbf{y})$ is nonlinear, its first-order approximation is affine. With non-negative weights, the complete problem is a convex QP.

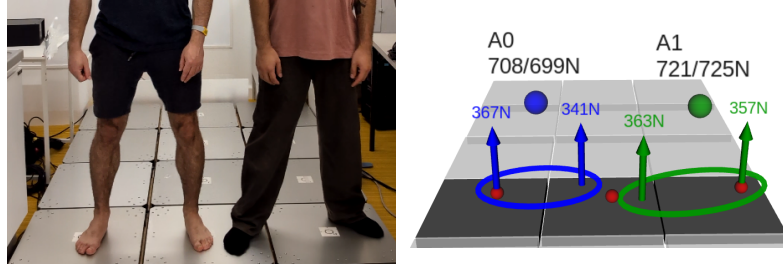


Figure 2.10: Recovered footprint positions and magnitudes after the force-share QP and sub-grid position QP during a measurement superposition, with the original sensor readings shown in red.

2.3.7 Cross-frame force sustainment

Short increases in measured force are common during gait, impacts, and load transfer and should not immediately be interpreted as evidence of a new agent. The force-share QP therefore reacts only when additional force remains near an existing agent for a sufficient duration.

For agent a , the total force currently available through its claims is

$$F_{a,k}^{\text{claim}} = F_{a,k}^{\text{fix}} + \sum_{p \in \mathcal{P}} C_{ap,k} F_{p,k}.$$

This quantity is evaluated before ambiguous footprint ownership is resolved. It is therefore independent of how the current QP distributes the force. This is important because otherwise the solver could suppress the apparent excess by assigning part of the footprint to the unassigned remainder.

The minimum additional force considered significant is

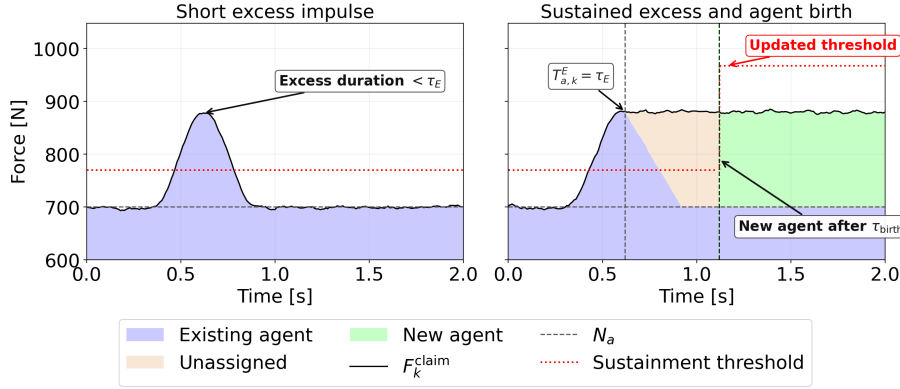


Figure 2.11: Force-sustainment behavior. A short excess does not remain above $N_a + \delta_{a,k}^E$ for the required duration τ_E and therefore does not activate sustainment. A sustained excess activates γ_a after τ_E , after which the multiplier increases while the excess persists.

$$\delta_{a,k}^E = \max(F_{\min}, \eta_E N_a),$$

where N_a is the current assignment target and $\eta_E > 0$ is a relative excess threshold hyperparameter. Thus an excess must be large enough both in absolute terms and relative to the expected load of the agent.

An excess is present when

$$F_{a,k}^{\text{claim}} > N_a + \delta_{a,k}^E. \quad (2.19)$$

Let $T_{a,k}^E$ denote how long this condition has held continuously. Its update is

$$T_{a,k}^E = \begin{cases} T_{a,k-1}^E + \Delta t_s, & \text{if eq. (2.19) holds,} \\ 0, & \text{otherwise.} \end{cases}$$

A short force impulse therefore disappears before the timer reaches the activation duration $\tau_E \geq 0$ and has no effect on the optimization. If the excess persists beyond τ_E , the sustained-excess multiplier becomes active and increases with the duration of the excess:

$$\gamma_{a,k} = \begin{cases} 0, & T_{a,k}^E < \tau_E, \\ \min\left(\gamma_{\max}, \gamma_0 + \frac{\gamma_{\max} - \gamma_0}{\tau_E^{\text{ramp}}} (T_{a,k}^E - \tau_E)\right), & T_{a,k}^E \geq \tau_E. \end{cases}$$

Here $0 \leq \gamma_0 \leq \gamma_{\max}$ sets the initial and maximum penalties after activation, and $\tau_E^{\text{ramp}} > 0$ determines the ramp duration. γ_0 , $\gamma_{\max}$, and τ_E^{ramp} are hyperparameters. Because it depends only on claims and previous timer state, the resulting $\gamma_{a,k}$ is computed before and held fixed during the current force-share QP. It weights the sustained-excess cost in the complete objective eq. (2.16). As the unexplained force remains present, γ_a increasingly discourages assigning that force to the existing agent. Where an unassigned share is admissible, the solver can instead retain part of the footprint as u_p , allowing it to enter the blob and agent-birth pipeline described in section 2.3.1.

Because the sustainment condition uses the claimed force before allocation, creating an unassigned residual does not remove the condition that activated it. No additional

release-hold state is therefore required. Once the excess disappears, $T_{a,k}^E$ resets and γ_a returns to zero. The impulse and sustained cases are compared in fig. 2.11.

2.3.8 Assignment application and persistent agent state

After all components are solved, $X_{ap,k}$ denotes the applied ownership share over the complete current footprint set: it equals the QP result for a contested pair after the thresholding and renormalization in section 2.3.6, one for a direct unambiguous assignment, and zero otherwise. Define the applied force contribution

$$F_{ap,k} := X_{ap,k} F_{p,k}.$$

Let $F_{\min}^{\text{ass}} > 0$ be the minimum assigned-force threshold used after the solve. It is an independent hyperparameter, distinct from the plate activation threshold $F_{\min} = 15 \text{ N}$, such that $F_{\min}^{\text{ass}} \neq F_{\min}$. The footprints available to agent a after assignment are defined once as

$$\mathcal{P}_{a,k}^{\text{ass}} := \{p \in \mathcal{P}_k : F_{ap,k} > F_{\min}^{\text{ass}}\}. \quad (2.20)$$

This post-solve threshold determines which assigned footprint portions are used by subsequent agent-level computations; it does not modify the exact force accounting of the force-share QP.

The associated resolved position for each $p \in \mathcal{P}_{a,k}^{\text{ass}}$ is

$$\mathbf{y}_{ap,k} = \begin{cases} \mathbf{y}_{ap,k}^{\text{sol}}, & \text{if footprint } p \text{ is shared,} \\ \mathbf{r}_{p,k}, & \text{if footprint } p \text{ is directly assigned.} \end{cases}$$

The corresponding measured agent force is

$$F_{a,k}^{\text{meas}} = \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} F_{ap,k}.$$

Whenever $\mathcal{P}_{a,k}^{\text{ass}} \neq \emptyset$, the assignment provides a valid force-weighted position measurement,

$$\mathbf{r}_{a,k}^{\text{meas}} = \frac{\sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} F_{ap,k} \mathbf{y}_{ap,k}}{F_{a,k}^{\text{meas}}}.$$

A valid measurement corrects the predicted state from section 2.3.3 using a linear Kalman filter [19]. The measurement vector and innovation are

$$\mathbf{y}_{a,k}^A = [(\mathbf{r}_{a,k}^{\text{meas}})^\top \quad F_{a,k}^{\text{meas}}]^\top, \quad \boldsymbol{\nu}_{a,k}^A = \mathbf{y}_{a,k}^A - \mathbf{H}^A \mathbf{x}_{a,k}^-.$$

Let $\tau_{\text{full}} \in (0, 1]$ be the full-assignment threshold, treated as a hyperparameter chosen close to unity, typically in the upper range of admissible ownership shares. The fractional- and trusted-assignment indicators are defined as

$$I_{\text{frac},a,k} := \mathbb{I}[\exists p \in \mathcal{P}_{a,k}^{\text{ass}} : X_{ap,k} < \tau_{\text{full}}], \\ I_{\text{trust},a,k} := \mathbb{I}[\forall p \in \mathcal{P}_{a,k}^{\text{ass}} : X_{ap,k} \geq \tau_{\text{full}}].$$

For a valid assignment, these conditions are complementary,

$$I_{\text{trust},a,k} = 1 - I_{\text{frac},a,k}.$$

The corresponding measurement covariance is

$$\mathbf{V}_{a,k}^A = \mathbf{V}_0^A + I_{\text{frac},a,k} \mathbf{V}_{\text{frac}}^A + \mathbb{I}[F_{a,k}^{\text{meas}} < F_{\min}^{\text{CoP}}] \mathbf{V}_{\text{low}}^A.$$

Here, $\mathbf{V}_0^A \succ \mathbf{0}$ is the nominal measurement covariance, $\mathbf{V}_{\text{frac}}^A \succeq \mathbf{0}$ increases the uncertainty associated with fractional assignments, while $\mathbf{V}_{\text{low}}^A \succeq \mathbf{0}$ increases the position uncertainty when the measured force is below the CoP reliability threshold $F_{\min}^{\text{CoP}}$.

Trusted measurements and agent dynamics profile

A valid measurement is trusted when $I_{\text{trust},a,k} = 1$, meaning that every contributing footprint has an ownership share at or above τ_{full} . Conversely, if any contributing footprint falls below this threshold, the assignment is fractional and $I_{\text{frac},a,k} = 1$. The threshold is chosen near unity so that only largely unambiguous assignments are treated as trusted.

For a trusted frame, define

$$F_{a,k}^{\text{trusted}} := F_{a,k}^{\text{meas}}.$$

The trusted force measurement adapts the agent's neutral force estimate using an exponential moving average with smoothing factor $\alpha_0 \in (0, 1]$:

$$F_a^0 \leftarrow (1 - \alpha_0)F_a^0 + \alpha_0 F_{a,k}^{\text{trusted}}. \quad (2.21)$$

The agent dynamics profile is subsequently updated with cadence, stance, velocity, acceleration, impulse, stride, spatial spread, and relative force bounds. Human, legged, wheeled, static-object, and custom profiles map these quantities to the claim radii and soft force envelope used in the following frame. For example, a learned maximum human stride expands σ_a^{stride} , whereas a static-object profile keeps the claim radius near its measured support extent. The neutral-force adaptation during running is illustrated in fig. 2.12.

2.3.9 Force-weighted influence field

For downstream proximity and interaction reasoning, each agent is represented by a compact force-weighted anisotropic influence field. Its center and total force are given by the filtered agent state, while its shape is estimated from the currently assigned positive-force footprint portions.

The resulting eight-scalar field is

$$\mathcal{F}_{a,k} = \left(\mathbf{r}_{a,k}, F_{a,k}, \sigma_{a,k}^x, \sigma_{a,k}^y, \varrho_{a,k}, \zeta_{a,k}^x, \zeta_{a,k}^y \right),$$

where $\mathbf{r}_{a,k} \in \mathbb{R}^2$ and $F_{a,k}$ are respectively the position and force of the persistent agent state, $\sigma_{a,k}^x$ and $\sigma_{a,k}^y$ describe the spatial extent of its support, $\varrho_{a,k}$ describes spatial correlation, and $\zeta_{a,k}^x, \zeta_{a,k}^y$ describe axis-wise directional skew.

The representation is motivated by limitations of a covariance ellipse. An unweighted covariance represents only the spatial distribution of support locations and gives the same importance to lightly and heavily loaded footprints. Force weighting instead

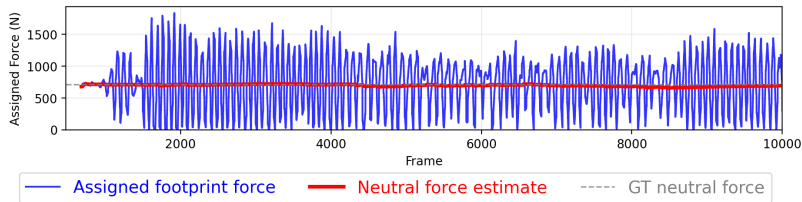


Figure 2.12: Estimated neutral force of a human subject while running.

makes strongly loaded contacts contribute more strongly to the estimated shape. However, normalized force weights remove absolute force magnitude, which is why the agent force $F_{a,k}$ is retained explicitly as part of the field. Furthermore, covariance is a second-order, centrally symmetric representation and cannot encode asymmetric support. Standardized third moments are therefore included to describe directional skew. These differences are illustrated in fig. 2.13.

The field shape is constructed from the common post-solve footprint set $\mathcal{P}_{a,k}^{\text{ass}}$ defined in eq. (2.20). When this set is non-empty, the normalized force weights are

$$w_{ap,k}^{\text{field}} = \frac{F_{ap,k}}{F_{a,k}^{\text{meas}}}, \quad \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k}^{\text{field}} = 1.$$

The footprint offsets are measured relative to the current agent position,

$$\mathbf{d}_{ap,k} = \mathbf{y}_{ap,k} - \mathbf{r}_{a,k}.$$

The force-weighted second moment about the agent position is then

$$\Sigma_{a,k}^{\text{meas}} = \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k}^{\text{field}} \mathbf{d}_{ap,k} \mathbf{d}_{ap,k}^{\top}.$$

Because $\mathbf{r}_{a,k}$ is supplied by the filtered agent state, this quantity is formally a force-weighted second moment about the tracked agent position.

Its component scales are

$$\sigma_{a,k}^{\nu,\text{meas}} = \sqrt{\max([\Sigma_{a,k}^{\text{meas}}]_{\nu\nu}, 0)}, \quad \nu \in \{x, y\},$$

Let $0 < \varepsilon_{\sigma} \ll 1$ be a length tolerance used in the subsequent standardization and correlation calculations. The corresponding correlation is

$$\varrho_{a,k}^{\text{meas}} = \begin{cases} \frac{[\Sigma_{a,k}^{\text{meas}}]_{xy}}{\sigma_{a,k}^{x,\text{meas}} \sigma_{a,k}^{y,\text{meas}}}, & \sigma_{a,k}^{x,\text{meas}} \sigma_{a,k}^{y,\text{meas}} > \varepsilon_{\sigma}^2, \\ 0, & \text{otherwise.} \end{cases}$$

The force-weighted standardized third moments are

$$\begin{aligned} \bar{d}_{ap,\nu,k} &= \frac{[\mathbf{d}_{ap,k}]_{\nu}}{\sigma_{a,k}^{\nu,\text{meas}} + \varepsilon_{\sigma}}, \\ \zeta_{a,k}^{\nu,\text{meas}} &= \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k}^{\text{field}} \bar{d}_{ap,\nu,k}^3, \quad \nu \in \{x, y\}. \end{aligned} \quad (2.22)$$

Collect the two components as $\zeta_{a,k}^{\text{meas}} := [\zeta_{a,k}^{x,\text{meas}} \ \zeta_{a,k}^{y,\text{meas}}]^{\top}$.

These quantities describe marginal skew along the world-frame x and y axes. They provide a compact, fixed-size indication of asymmetric support; the complete third-order spatial moment tensor is not estimated.

The force weighting causes heavily loaded footprint portions to influence the field shape more strongly than lightly loaded portions at the same distance. Multiplying every assigned footprint force by a common factor $c_F > 0$ does not change the normalized weights or the resulting shape parameters. Absolute loading information is instead supplied directly by the agent-state force $F_{a,k}$.

When $\mathcal{P}_{a,k}^{\text{ass}} = \emptyset$, no reliable instantaneous shape measurement is available and the previous field shape is retained.

For a robot or object with known contact geometry, fitted model contacts may replace the measured shape estimate before temporal smoothing. This modifies only

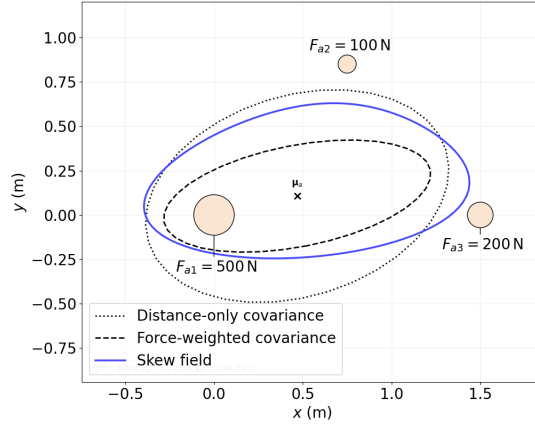


Figure 2.13: Comparison of a symmetric covariance field and the force-weighted skew field used for interaction reasoning.

the shape parameters; the field center remains $\mathbf{r}_{a,k}$ and the field force remains $F_{a,k}$. The model can therefore refine the expected physical support geometry without replacing the state measured by the floor.

Let the selected target shape after optional model fitting be represented by

$$\Sigma_{a,k}^{\text{target}}, \quad \zeta_{a,k}^{\text{target}}.$$

Without a model prior, $\Sigma_{a,k}^{\text{target}} := \Sigma_{a,k}^{\text{meas}}$ and $\zeta_{a,k}^{\text{target}} := \zeta_{a,k}^{\text{meas}}$. If no instantaneous shape measurement is available, both targets are set to $\Sigma_{a,k-1}$ and $\zeta_{a,k-1}$, respectively. At agent birth, the field is initialized with $\Sigma_{a,0} = ((d_a^{\min})^2/4)\mathbf{I}_2$ and $\zeta_{a,0} = \mathbf{0}$.

To suppress frame-to-frame fluctuations, these shape quantities are exponentially smoothed with gain $\alpha_{\text{shape}} \in [0, 1]$,

$$\begin{aligned} \tilde{\Sigma}_{a,k} &= (1 - \alpha_{\text{shape}})\Sigma_{a,k-1} + \alpha_{\text{shape}}\Sigma_{a,k}^{\text{target}}, \\ \tilde{\zeta}_{a,k} &= (1 - \alpha_{\text{shape}})\zeta_{a,k-1} + \alpha_{\text{shape}}\zeta_{a,k}^{\text{target}}. \end{aligned} \quad (2.23)$$

Only the field shape is smoothed here. The center and force already belong to the persistent filtered agent state and are therefore used directly without introducing an additional influence-field filter.

A minimum physical field diameter $d_a^{\min} > 0$ is enforced on the smoothed second moment, and $0 < \varepsilon_{\Sigma} \ll 1$ supplies a numerical lower bound on its principal scales. After symmetrization,

$$\frac{\tilde{\Sigma}_{a,k} + \tilde{\Sigma}_{a,k}^{\top}}{2} = \mathbf{U}_{a,k} \text{diag}(\lambda_{a,1,k}^{\Sigma}, \lambda_{a,2,k}^{\Sigma}) \mathbf{U}_{a,k}^{\top}.$$

Here $\mathbf{U}_{a,k}$ is orthogonal and $\lambda_{a,1,k}^{\Sigma}, \lambda_{a,2,k}^{\Sigma}$ are the eigenvalues of the symmetrized matrix.

The effective eigenvalues are

$$\lambda_{a,i,k}^{\Sigma, \text{eff}} = \max\left(\lambda_{a,i,k}^{\Sigma}, \frac{(d_a^{\min})^2}{4}, \varepsilon_{\Sigma}^2\right), \quad i \in \{1, 2\},$$

giving

$$\Sigma_{a,k} = \mathbf{U}_{a,k} \text{diag} \left(\lambda_{a,1,k}^{\Sigma,\text{eff}}, \lambda_{a,2,k}^{\Sigma,\text{eff}} \right) \mathbf{U}_{a,k}^{\top}.$$

The final spatial scales are recovered from this matrix,

$$\sigma_{a,k}^x = \sqrt{[\Sigma_{a,k}]_{xx}}, \quad \sigma_{a,k}^y = \sqrt{[\Sigma_{a,k}]_{yy}},$$

with correlation

$$\varrho_{a,k} = \frac{[\Sigma_{a,k}]_{xy}}{\sigma_{a,k}^x \sigma_{a,k}^y}.$$

The minimum eigenvalue is strictly positive, so $\Sigma_{a,k}$ is positive definite and invertible. For a query point $\mathbf{r}$, its displacement from the agent is

$$\mathbf{d} = \mathbf{r} - \mathbf{r}_{a,k}, \quad \bar{d}_{\nu} = \frac{d_{\nu}}{\sigma_{a,k}^{\nu} + \varepsilon_{\sigma}}, \quad \nu \in \{x, y\}.$$

The symmetric component of the field is the squared Mahalanobis score [27]

$$\delta_{\text{sym},a,k}^2(\mathbf{r}) = \mathbf{d}^{\top} \Sigma_{a,k}^{-1} \mathbf{d}.$$

Directional asymmetry is introduced through the bounded deformation

$$b_{a,k}(\mathbf{r}) = \beta_x^{\text{skew}} \tanh(\zeta_{a,k}^x) \tanh(\bar{d}_x) + \beta_y^{\text{skew}} \tanh(\zeta_{a,k}^y) \tanh(\bar{d}_y).$$

Here $\beta_x^{\text{skew}}, \beta_y^{\text{skew}} \geq 0$ are fixed dimensionless deformation gains.

The resulting skew-aware field score is

$$\delta_{\text{field},a,k}^2(\mathbf{r}) = \delta_{\text{sym},a,k}^2(\mathbf{r}) \exp(-b_{a,k}(\mathbf{r})).$$

Because $b_{a,k}(\mathbf{r})$ is bounded, the skew term can stretch or contract the field directionally without overcoming the quadratic growth of the Mahalanobis component. Consequently,

$$\delta_{\text{field},a,k}^2(\mathbf{r}) \rightarrow \infty \quad \text{as} \quad \|\mathbf{r} - \mathbf{r}_{a,k}\|_2 \rightarrow \infty.$$

The resulting quantity is a skew-aware influence score with no probabilistic or metric interpretation. With zero skew it reduces exactly to the covariance-based squared Mahalanobis field. Non-zero axis-wise third moments deform the otherwise elliptical contours according to asymmetric force support, while the center and force remain exactly those of the persistent agent state. The representation therefore adds a compact description of support geometry without introducing redundant estimates of agent position or force.

2.4 Post-tracking applications

2.4.1 Scene- and model-informed tracking

The tracking stages in section 2.3 require only force-floor measurements. Additional scene information can reduce ambiguity when properties of an entity are known. A scene description may provide an expected neutral force, a dynamics class, physical extent, static-state information, or a known initial pose. These quantities refine the force envelopes, reach limits, and motion priors used by the claim and assignment stages. Human profiles retain the wider force and stride variation required by gait, while wheeled, legged, and static entities can use bounds matched to their expected dynamics.

For articulated robots or objects with known geometry, ground-contact regions are extracted from the Unified Robot Description Format (URDF) model [28] and represented by planar contact centers and support radii. The contact partitions defined in section 2.4.2 provide the layout corresponding to the current number of supported contacts and supply preferred relative positions to the sub-grid position solver. The same geometry can provide a minimum spatial extent or reference second-moment matrix for the force-weighted influence field. Contact-count-specific layouts allow the model prior to adapt when only a subset of feet or wheels supports the body.

Scene and model terms are therefore additional priors on the same measured footprint assignments: measured force, first moments, and plate-support constraints remain the physical quantities that determine the accepted state.

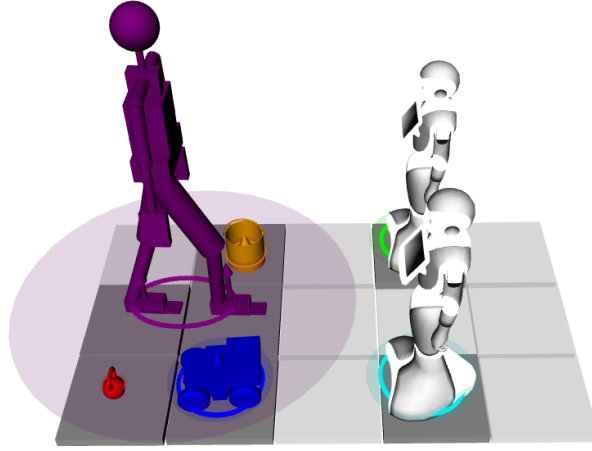


Figure 2.14: Scene-informed tracking and model fusion, with adjusted claim radii and influence fields.

2.4.2 Partition-based robot pose fusion

This stage assumes that a robot model has already been associated with tracked agent a . The possible ground-contact configurations of the model are compared with the resolved footprint portions assigned to that agent. The configuration that best matches the observed footprints is then used to estimate the planar position and heading of the robot model.

Model-contact partitions. Let

$$\mathcal{C}_a = \{\mathbf{c}_1, \dots, \mathbf{c}_M\}, \quad \mathbf{c}_m = \mathcal{M}_a \mathbf{r}_{M_a C_m},$$

denote the planar ground-contact centers obtained from the robot model, where M is the number of modeled contact centers. Because several nearby model contacts may appear on the force floor as a single footprint, the model contacts are grouped into set partitions

$$\Pi = \{G_1, \dots, G_q\}, \quad \bigcup_{j=1}^q G_j = \{1, \dots, M\}, \quad G_j \cap G_{j'} = \emptyset \quad (j \neq j'),$$

where each group G_j contains the model contacts that may be measured as one force-floor footprint.

Each group is represented by the geometric center of its contacts,

$$\boldsymbol{\mu}_j^\Pi = \frac{1}{|G_j|} \sum_{m \in G_j} \mathbf{c}_m, \quad j = 1, \dots, q.$$

All set partitions are generated once from the model geometry. A partition is discarded if two group centers coincide or if the diameter of any group exceeds the diagonal of one force plate, because that group cannot contribute to a single plate measurement. The remaining partitions with q groups form the candidate set $\mathfrak{P}_{a,q}$.

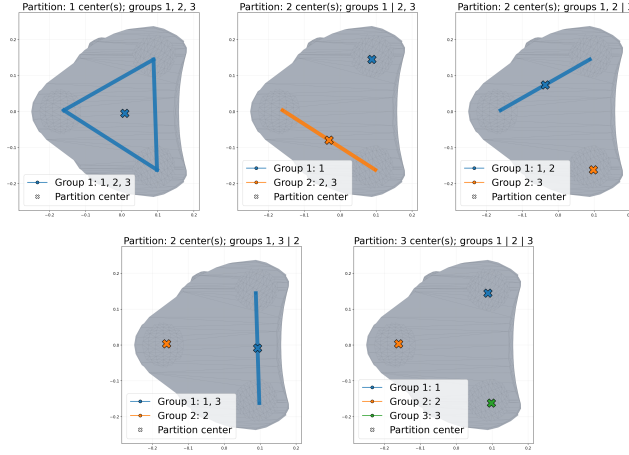


Figure 2.15: Example model-contact partitions for the Pepper robot with three observed ground contacts.

Matching to assigned footprints. Pose estimation uses the common post-solve footprint set $\mathcal{P}_{a,k}^{\text{ass}}$ from eq. (2.20), together with the resolved positions from section 2.3.6. Define

$$\mathcal{P}_{a,k}^{\text{pose}} = \{p \in \mathcal{P}_{a,k}^{\text{ass}} : F_{p,k} > F_{\min}^{\text{CoP}}\}, \quad q_{a,k} = |\mathcal{P}_{a,k}^{\text{pose}}|.$$

Here, $q_{a,k}$ is the number of available footprints assigned to agent a at frame k , whose parent footprint's force exceeds the minimum threshold for a reliable CoP. Their positions $\mathbf{y}_{ap,k}$ and forces $F_{ap,k}$ are already known after assignment application. Only partitions with the same number of groups, $\Pi \in \mathfrak{P}_{a,q_{a,k}}$, are considered. If no valid partition exists for the current assigned-footprint count, the pose fit is treated as unavailable and the fallback described below is used.

Each footprint is weighted according to its assigned force and share. With $\kappa_{\text{full}} = 10$ and the full-assignment threshold τ_{full} introduced in section 2.3.8, define

$$w_{ap,k} = F_{ap,k} [1 + (\kappa_{\text{full}} - 1) \mathbb{I}[X_{ap,k} \geq \tau_{\text{full}}]].$$

Therefore, footprints carrying more force have a greater influence on the fit. A footprint assigned entirely to agent a receives additional weight because its position is not influenced by ambiguous force sharing.

The resulting heading estimates are illustrated in fig. 2.16.

Optimal partition correspondence. For each candidate partition, the previous robot heading provides an approximate world-frame placement of the group centers.

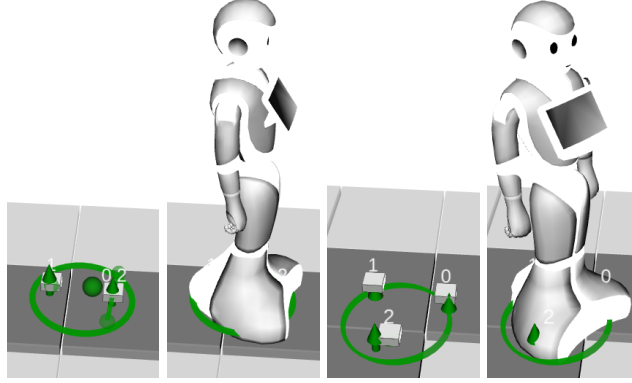


Figure 2.16: Heading of the Pepper robot model estimated from the resolved footprint positions and fitted model geometry.

The geometric centers of the measured footprints and of the partition are

$$\bar{\mathbf{y}}_{a,k} = \frac{1}{q_{a,k}} \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} \mathbf{y}_{ap,k}, \quad \bar{\boldsymbol{\mu}}_{\Pi} = \frac{1}{q_{a,k}} \sum_{j=1}^{q_{a,k}} \boldsymbol{\mu}_j^{\Pi}.$$

The approximate world-frame position of partition center j is then

$$\mathbf{y}_{j,k}^{\Pi, \text{prior}} = \bar{\mathbf{y}}_{a,k} + \mathbf{R}(\theta_{a,k-1}) (\boldsymbol{\mu}_j^{\Pi} - \bar{\boldsymbol{\mu}}_{\Pi}).$$

Thus, the candidate partition is rotated using the previous robot heading and translated so that its center coincides with the center of the observed footprints. The partition centers are then matched one-to-one with the observed footprint positions by minimizing the weighted squared distance,

$$\pi_{\Pi} \in \arg \min_{\pi \in \mathfrak{S}_{a,k}^{\text{pose}}} \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} \left\| \mathbf{y}_{ap,k} - \mathbf{y}_{\pi(p),k}^{\Pi, \text{prior}} \right\|_2^2,$$

where $\mathfrak{S}_{a,k}^{\text{pose}}$ is the set of bijections from $\mathcal{P}_{a,k}^{\text{ass}}$ to $\{1, \dots, q_{a,k}\}$. The optimal assignment is found using the Hungarian algorithm [29], ensuring that each footprint is matched to exactly one model group. The prior heading is initialized from a known association pose or, when the agent is moving, from its velocity direction. If neither is available at the first fit, every bijection is evaluated once and the lowest-error result initializes the heading.

Weighted rigid registration. For $q_{a,k} \geq 2$, after determining the correspondence π_{Π} , each matched model center $\boldsymbol{\mu}_{\pi_{\Pi}(p)}^{\Pi}$ receives the weight $w_{ap,k}$ of its corresponding footprint. The rigid registration then finds the rotation and translation that best align the model contacts with the measured footprints,

$$(\theta_{\Pi,k}^{\text{fit}}, \mathbf{t}_{\Pi,k}^{\text{fit}}) \in \arg \min_{\theta, \mathbf{t}} \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} \left\| \mathbf{y}_{ap,k} - \mathbf{t} - \mathbf{R}(\theta) \boldsymbol{\mu}_{\pi_{\Pi}(p)}^{\Pi} \right\|_2^2. \quad (2.24)$$

The weighted center of the measured footprints is

$$\bar{\mathbf{y}}_{a,k}^w = \frac{\sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} \mathbf{y}_{ap,k}}{\sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k}},$$

and the weighted center of the matched model partition is

$$\bar{\boldsymbol{\mu}}_{\Pi}^w = \frac{\sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} \boldsymbol{\mu}_{\pi_{\Pi}(p)}^{\Pi}}{\sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k}}.$$

The model centers therefore have no predefined weights. After matching, each model center receives the weight of its corresponding footprint.

Define the centered positions as

$$\mathbf{y}'_{ap,k} = \mathbf{y}_{ap,k} - \bar{\mathbf{y}}_{a,k}^w, \quad \boldsymbol{\mu}'_{p,\Pi} = \boldsymbol{\mu}_{\pi_{\Pi}(p)}^{\Pi} - \bar{\boldsymbol{\mu}}_{\Pi}^w.$$

Writing

$$\mathbf{y}'_{ap,k} = \begin{bmatrix} y'_{ap,k,x} \\ y'_{ap,k,y} \end{bmatrix}, \quad \boldsymbol{\mu}'_{p,\Pi} = \begin{bmatrix} \mu'_{p,\Pi,x} \\ \mu'_{p,\Pi,y} \end{bmatrix},$$

the rotation that gives the best planar alignment is obtained from

$$A_{\Pi,k} = \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} (\mu'_{p,\Pi,x} y'_{ap,k,y} - \mu'_{p,\Pi,y} y'_{ap,k,x}),$$

$$B_{\Pi,k} = \sum_{p \in \mathcal{P}_{a,k}^{\text{ass}}} w_{ap,k} (\mu'_{p,\Pi,x} y'_{ap,k,x} + \mu'_{p,\Pi,y} y'_{ap,k,y}),$$

giving

$$\theta_{\Pi,k}^{\text{fit}} = \text{atan2}(A_{\Pi,k}, B_{\Pi,k}).$$

The corresponding translation is

$$\mathbf{t}_{\Pi,k}^{\text{fit}} = \bar{\mathbf{y}}_{a,k}^w - \mathbf{R}(\theta_{\Pi,k}^{\text{fit}}) \bar{\boldsymbol{\mu}}_{\Pi}^w.$$

Let $E_{\Pi,k}$ be the minimized objective value in eq. (2.24), and let Π_k^* be the candidate with the smallest value. Candidates within the fixed tolerance $\varepsilon_E > 0$, which has the same units as $E_{\Pi,k}$, are treated as tied with the minimum. Among them, the candidate with the smallest wrapped heading change from the previous frame is selected.

The selected pose is smoothed across frames with $\alpha_{\text{pose}} \in (0, 1]$. Using $\text{wrap}_{\pi}(\cdot)$ to wrap an angle to $(-\pi, \pi]$, the accepted model pose is

$$\theta_{a,k} = \theta_{a,k-1} + \alpha_{\text{pose}} \text{wrap}_{\pi}(\theta_{\Pi_k^*,k}^{\text{fit}} - \theta_{a,k-1}),$$

$$\mathbf{t}_{a,k}^{\text{model}} = (1 - \alpha_{\text{pose}}) \mathbf{t}_{a,k-1}^{\text{model}} + \alpha_{\text{pose}} \mathbf{t}_{\Pi_k^*,k}^{\text{fit}}.$$

If only one usable footprint is available, its position provides information about translation but not heading, so the previous heading is retained and $\mathbf{t}_{a,k}^{\text{model}} = \mathbf{y}_{ap,k} - \mathbf{R}(\theta_{a,k-1}) \boldsymbol{\mu}_1^{\Pi}$. If no valid partition can be found, the previous heading is kept and the model position is set to the tracked agent position, $\mathbf{t}_{a,k}^{\text{model}} := \mathbf{r}_{a,k}$. The resulting robot-model pose is used by the three-dimensional geometric queries in sections 2.4.3 and 2.4.4.

If available, internal robot odometry can be fused with the force-floor pose estimate to improve tracking. The odometry is treated as a separate measurement with its own uncertainty, and the two pose estimates are combined using a Kalman filter, with the odometry being the sole source of heading information during single-plate support.

2.4.3 Touch-contact dynamics

Touch detection is performed only after the force shares and sub-grid footprint positions have been assigned. These assignments are assumed to be correct, so contact recognition does not modify ownership, force, or position. It uses only the resolved human footprint position and the fitted geometry of the other tracked agents. For robots, the fitted geometry is obtained from the pose-fusion procedure in section 2.4.2.

Each resolved human footprint position defines a vertical line through the world at that planar location. Let $p \in \mathcal{P}_{a,k}^{\text{ass}}$ be a footprint assigned to human a , with resolved planar position $\mathbf{y}_{ap,k} \in \mathbb{R}^2$, and let $\mathcal{G}_{d,k} \subset \mathbb{R}^3$ be the fitted model surface of another agent $d \in \mathcal{A}_k \setminus \{a\}$. The heights at which the vertical line through the footprint crosses that model are

$$\mathcal{Z}_{apd,k} = \left\{ z \geq 0 : \begin{bmatrix} \mathbf{y}_{ap,k} \\ z \end{bmatrix} \in \mathcal{G}_{d,k} \right\}.$$

The method records a footprint–model contact when $\mathcal{Z}_{apd,k} \neq \emptyset$. If the line crosses one or more models at several heights, the reported target and height are selected together as

$$(d^*, z^*) \in \arg \max_{\substack{d \in \mathcal{A}_k \setminus \{a\}, \\ z \in \mathcal{Z}_{apd,k}}} z.$$

Equal-height ties are resolved by the agent identifier. The exact world-frame contact position is therefore

$$\mathbf{r}_{ap,k}^{\text{contact}} = \begin{bmatrix} \mathbf{y}_{ap,k} \\ z^* \end{bmatrix}.$$

Thus the assigned footprint supplies the two horizontal coordinates, while the model intersection supplies the vertical coordinate. If every $\mathcal{Z}_{apd,k}$ is empty, the footprint remains a ground contact only. The reconstruction is illustrated in fig. 2.17.

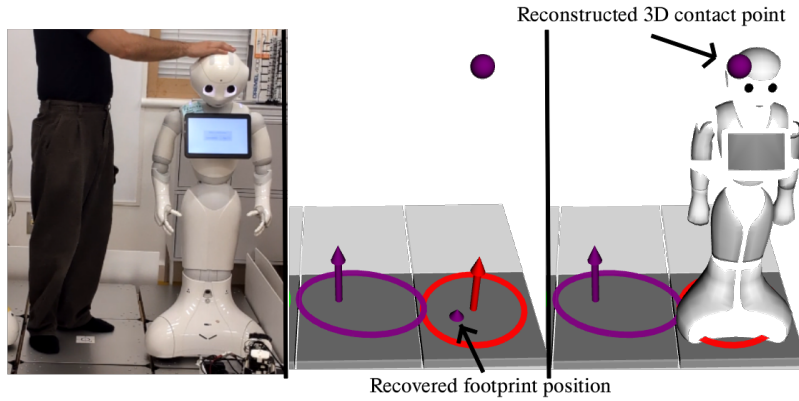


Figure 2.17: Touch-contact detection using a resolved human footprint and the fitted geometry of a Pepper robot. The vertical intersection provides the contact height.

2.4.4 Carrying dynamics

Carrying is represented as an explicit directed relation from a carried object d to its carrier a . The relation changes how force is explained and how the object state evolves, but does not merge their semantic identities. Only object agents tagged

as carryable may serve as d . Let $\delta_{\text{abs}}^F > 0$ and $\eta_{\text{rel}}^F > 0$ be the absolute and relative force-transfer tolerance hyperparameters, and define

$$\delta_d^F := \max(\delta_{\text{abs}}^F, \eta_{\text{rel}}^F F_d^0).$$

Attachment. Two attachment routes are used. The first detects a human pickup of an existing static object. While the object is stably supported, the tracker stores its supporting footprint set $\mathcal{P}_d^{\text{supp}}$ and its baseline force F_d^{supp} . Treating an inactive footprint as zero force, the lost support force is

$$F_{d,k}^{\text{loss}} := \max\left(0, F_d^{\text{supp}} - \sum_{p \in \mathcal{P}_d^{\text{supp}}} F_{p,k}\right).$$

A pickup candidate is created when $|F_{d,k}^{\text{loss}} - F_d^0| \leq \delta_d^F$ continuously for the attachment duration τ_{attach} . If the object is not already carried and an eligible human is within the attachment radius r_{attach} , the nearest human becomes the carrier, with agent identifier resolving an equal-distance tie. This preserves the existing object track after its direct floor support disappears.

The second route applies when a modeled object lies on a valid support surface of another modeled, non-human agent. The support relation is detected using the same model-intersection procedure described in section 2.4.3. In both cases, the carried object keeps its identifier, neutral force, and the source-footprint identifiers associated with the attachment transition.

Force ownership while carried. As defined in eq. (2.7), the neutral forces of all $d \in \mathcal{D}_a^{\text{carry}}$ are included in the carrier target N_a . A carried object does not claim force-floor footprints independently. Instead, the force-share QP assigns the combined load to the carrier against N_a . The neutral-force estimates and dynamics profiles of both participants are not relearned from the combined measurement. A single non-solver footprint is retained only to propagate the carried object’s semantic position. It stores F_d^0 but does not introduce an additional measured-force owner.

Carried pose. For a carrier with a corresponding 3D model and an available heading $\theta_{a,k}$, the planar offset of the object is stored in the carrier frame at attachment. Robot headings come from section 2.4.2.

$$\mathbf{o}_{ad} = \mathbf{R}(\theta_{a,k_0})^\top (\mathbf{r}_{d,k_0} - \mathbf{r}_{a,k_0}),$$

where k_0 is the attachment frame. The object’s planar position subsequently follows the carrier according to

$$\mathbf{y}_{d,k}^{\text{carry}} = \mathbf{r}_{a,k} + \mathbf{R}(\theta_{a,k})\mathbf{o}_{ad}.$$

The vertical coordinate is reconstructed from the carrier geometry using the same procedure as for touch-contact reconstruction. Let $\mathcal{G}_{a,k}$ be the fitted surface of the carrier model. The possible support heights above the propagated planar position are

$$\mathcal{Z}_{ad,k}^{\text{carry}} = \left\{ z \geq 0 : \begin{bmatrix} \mathbf{y}_{d,k}^{\text{carry}} \\ z \end{bmatrix} \in \mathcal{G}_{a,k} \right\}.$$

When this set is non-empty, the carried object’s model position is placed on the highest intersected support surface,

$$z_{d,k}^{\text{carry}} = \max \mathcal{Z}_{ad,k}^{\text{carry}},$$

$$\mathbf{r}_{d,k}^{\text{model}} = \begin{bmatrix} \mathbf{y}_{d,k}^{\text{carry}} \\ z_{d,k}^{\text{carry}} \end{bmatrix}.$$

Thus the stored carrier-relative offset determines where the object moves in the horizontal plane, while the fitted carrier model determines its vertical support height. This is the same geometric reconstruction used for touch detection, applied here persistently while the carry relation remains active.

Whenever the carrier model or heading is unavailable, no support-surface reconstruction is available and the carried object inherits the carrier position directly,

$$\mathbf{r}_{d,k} := \mathbf{r}_{a,k}.$$

Thus no independent vertical coordinate is inferred for the carried object.

Release. A release candidate must be a footprint in the carrier's common post-solve set $\mathcal{P}_{a,k}^{\text{ass}}$ whose assigned force matches the carried object's neutral force. Define

$$\mathcal{P}_{ad,k}^{\text{rel}} = \{p \in \mathcal{P}_{a,k}^{\text{ass}} : |F_{ap,k} - F_d^0| \leq \delta_d^F\}.$$

If several candidates exist, the closest force match is selected, followed by the smallest source-footprint identifier. The release duration is accumulated only while the same carrier and source footprint remain selected,

$$T_{d,k}^{\text{rel}} = \begin{cases} T_{d,k-1}^{\text{rel}} + \Delta t_s, & \mathcal{P}_{ad,k}^{\text{rel}} \neq \emptyset \text{ and } p_k^* = p_{k-1}^*, \\ \Delta t_s, & \mathcal{P}_{ad,k}^{\text{rel}} \neq \emptyset \text{ with a new candidate,} \\ 0, & \mathcal{P}_{ad,k}^{\text{rel}} = \emptyset. \end{cases}$$

Release is confirmed when $T_{d,k}^{\text{rel}} \geq \tau_{\text{rel}}$. The selected footprint is transferred from a to d , the object is snapped to its solved position and force, and the carry relation is cleared. Force sustainment is suppressed for both participants during this transition so the known transfer is not interpreted as an excess or deficit.

The object's model support is then evaluated again using the same geometric support test. It may attach to another valid modeled carrier or remain independent if no supporting surface exists. In the independent case, N_a no longer includes F_d^0 , and the ordinary force and position QPs resume separate ownership estimation for a and d in the following frame.

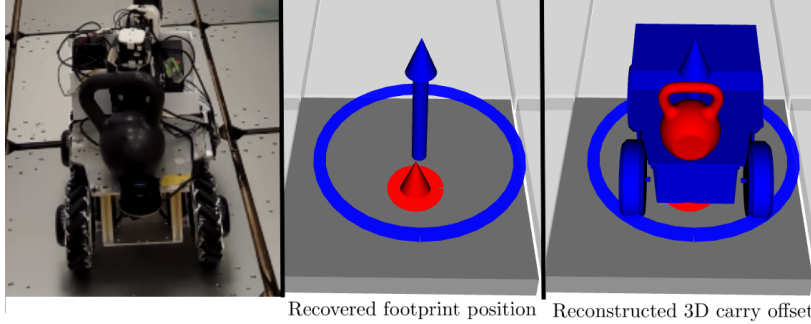


Figure 2.18: Wheeled rover carrying a kettlebell, with the three-dimensional position of the carried object reconstructed from the recovered footprint position and the carrier model geometry.

An example of the resulting carried-object pose is shown in fig. 2.18.

2.5 Simulation through the force-floor representation

Simulation is used to evaluate the tracking framework in environments larger and more densely populated than the physical experimental floor. The same force-plate representation is used in simulation, so the tracker receives the same type of input regardless of whether the contacts originate from physical hardware, human gait data, or a physics simulator.

2.5.1 Digital force-floor simulation

Each physical force plate from section 2.2.1 is reproduced by a digital clone with the same dimensions, local coordinate frame, and four corner-force channels. Because these plates are purely virtual, the same geometry can be tiled without the hardware limitations of the experimental setup, allowing simulated environments containing hundreds of plates.

At every simulation frame, each ground contact supplies a planar position $\mathbf{r}_{j,k}$ and vertical force $F_{j,k}$. The contact is first assigned to the virtual sensor $s \in S$ whose support contains its position. Multiple contacts falling on the same plate are combined into the footprint $p = p(s, k)$ before the measurement is passed to the tracker.

The simulator also constructs the four virtual corner-sensor measurements. This uses the same bilinear diagonal-response model introduced for the sub-grid position QP in section 2.3.6. For a contact contributing to footprint $p = p(s, k)$, let

$$(\xi_{j,k}, v_{j,k}) = \boldsymbol{\xi}_p(\mathbf{r}_{j,k}), \quad \xi_{j,k}, v_{j,k} \in [0, 1],$$

be its normalized plate coordinates. Its force contribution to the four corner sensors is

$$\begin{aligned} F_{p,1,j,k}^{\text{sim}} &= F_{j,k}(1 - \xi_{j,k})(1 - v_{j,k}), & F_{p,2,j,k}^{\text{sim}} &= F_{j,k}\xi_{j,k}(1 - v_{j,k}), \\ F_{p,3,j,k}^{\text{sim}} &= F_{j,k}\xi_{j,k}v_{j,k}, & F_{p,4,j,k}^{\text{sim}} &= F_{j,k}(1 - \xi_{j,k})v_{j,k}. \end{aligned}$$

The virtual reading of corner sensor ℓ is therefore

$$F_{p,\ell,k}^{\text{sim}} = \sum_{j \in \mathcal{J}_{p,k}} F_{p,\ell,j,k}^{\text{sim}}, \quad \ell \in \{1, 2, 3, 4\},$$

where $\mathcal{J}_{p,k}$ contains the simulated contacts contributing to footprint p .

This construction preserves the total contact force,

$$F_{p,k}^{\text{sim}} = \sum_{\ell=1}^4 F_{p,\ell,k}^{\text{sim}} = \sum_{j \in \mathcal{J}_{p,k}} F_{j,k},$$

while also reproducing the diagonal response used by the position solver,

$$\chi_{p,k}^{\text{sim}} = F_{p,1,k}^{\text{sim}} - F_{p,2,k}^{\text{sim}} + F_{p,3,k}^{\text{sim}} - F_{p,4,k}^{\text{sim}} = \sum_{j \in \mathcal{J}_{p,k}} F_{j,k}(1 - 2\xi_{j,k})(1 - 2v_{j,k}).$$

The simulated CoP is obtained from the resulting four corner channels using eqs. (2.2) and (2.3). Consequently, the simulated plate produces the same force, CoP, and diagonal information used by the physical tracking pipeline. The virtual measurements are emitted at the same sampling rate f_s .

2.5.2 Human simulation

Human motion is generated from the treadmill walking dataset of van der Zee *et al.* [20], which provides synchronized motion-capture trajectories and left- and right-foot ground-reaction forces.

Because the recordings describe treadmill walking, the recorded motion is converted into a continuously translating walking animation. The supporting foot is used as a temporary ground anchor while the marker trajectories are replayed, and the accumulated displacement is transferred to the character to produce forward translation. Successive support changes therefore produce continuous forward motion while retaining the recorded gait kinematics and vertical ground-reaction-force profiles. The resulting animation can be translated and rotated freely through the simulated environment while the original foot-force sequence is preserved.

Each active foot provides its current planar ground-contact position and recorded vertical force. These contacts are then passed directly to the digital force-floor model described in section 2.5.1.

2.5.3 Robot simulation

Robot motion is simulated in Isaac Lab [21]. The simulation contains heterogeneous legged and wheeled robot models and provides their physical interaction with the ground. For each active support contact, the simulator supplies the contact location and vertical ground-reaction force. These simulated forces are used directly as the contact inputs $\mathbf{r}_{j,k}, F_{j,k}$ to the virtual force floor. No tracker-specific force distribution is generated for the robots: the forces arise from the simulated robot dynamics and are only converted into virtual plate measurements by the common plate model.

2.5.4 Simulated trajectories

The motion of both humans and robots through their respective environments is controlled at the trajectory level using Optimal Reciprocal Collision Avoidance (ORCA) [22]. ORCA generates collision-avoiding planar velocity commands for multiple simultaneously moving agents.

For human simulation, the ORCA velocity determines the direction and speed at which the reconstructed walking animation progresses through the virtual floor. For robot simulation, the same trajectory layer supplies motion commands to the corresponding simulated robot controllers. ORCA therefore determines where the agents move, while the human gait recordings or Isaac dynamics independently determine the ground-contact positions and forces produced during that motion.

This separation allows the same crowd and crossing scenarios to be reproduced with either humans or heterogeneous robots while maintaining the same force-floor observation model at the tracker interface.

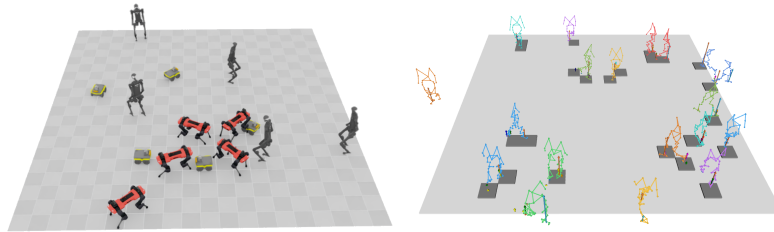


Figure 2.19: Isaac Lab simulation of heterogeneous robots (left) and human gait simulation (right) on a 20x20 virtual grid.

Chapter 3

Experimental Validation

This chapter validates the force-floor tracker against two independent references: optical motion capture (MoCap) and a precision force plate. The validation proceeds in four stages. First, the force plates are registered in the MoCap volume. Second, rigid marker frames, projected center-of-mass (CoM) locations, and reference weights are identified. Third, the intrinsic CoP accuracy of the force plates is characterized. Finally, the complete tracker is evaluated in solo and multi-agent episodes. The experiments involve a SoftBank Pepper robot [24], a Vstone MecanumRover 3 [25], two adult human participants, a 15.1 kg bucket, and an 8 kg kettlebell. Hereafter, *robots* refers to Pepper and the MecanumRover 3, *humans* to the two adult participants, and *objects* to the bucket and kettlebell.

Hardware and software specifications. The following experiments were conducted on a Dell G15 5515 laptop equipped with an AMD Ryzen 7 5800H processor and 32 GB of RAM [30, 31]. GPU acceleration was not used. The system ran Ubuntu 22.04.5 LTS [32] with ROS 2 Humble Hawksbill [33], and the tracking pipeline was implemented in Python 3.10.2 [34] as a single ROS 2 node without parallelization. Force-plate measurements were acquired using a custom C# desktop application and transmitted to the ROS 2 network through a custom TCP bridge. Motion-capture ground truth was obtained using an eight-camera OptiTrack Flex system operated with Motive 2.3 [35] and configured at 200 Hz on a Windows 11 desktop. Motion-capture trajectories were exported from Motive as CSV files, while the corresponding tracking data were recorded as ROS 2 bag files for offline analysis.

3.1 MoCap reference environment

3.1.1 Plate registration

An optical MoCap volume was installed around the complete instrumented floor. The calibration recovers the plate-pose quantities introduced in section 2.2.4: for each plate sensor $s \in S$, its origin ${}_{\mathcal{I}}\mathbf{r}_{IS_s}$ in the common world frame $\mathcal{I}$ and its planar orientation θ_s .

A custom L-shaped calibration target with three reflective markers was placed on each plate in turn. For each recording, marker positions were averaged over all valid MoCap frames to reduce measurement noise. The three markers were then identified geometrically as the target corner, right-arm marker, and top-arm marker. The right arm determines the plate yaw, while the top arm verifies the corresponding perpendicular target orientation. Let $\mathbf{m}_{s,C}$ and $\mathbf{m}_{s,R}$ denote the averaged planar world-frame positions of the corner and right-arm markers for plate s . The plate

yaw is

$$\theta_s = \text{atan2} \left(m_{s,R}^y - m_{s,C}^y, m_{s,R}^x - m_{s,C}^x \right). \quad (3.1)$$

Let $_{\mathcal{S}_s} \mathbf{d}_C$ denote the known vector from the plate origin to the target corner marker, expressed in the plate-local frame $\mathcal{S}_s$. Its magnitude is fixed by the calibration-target geometry, with $\|_{\mathcal{S}_s} \mathbf{d}_C\|_2 = 0.1765$ m. The plate origin is then

$${}_{\mathcal{I}} \mathbf{r}_{IS_s} = \mathbf{m}_{s,C} - \mathbf{R}(\theta_s)_{\mathcal{S}_s} \mathbf{d}_C. \quad (3.2)$$

Repeating the procedure for every $s \in S$ reconstructs the complete force-floor layout in $\mathcal{I}$. These calibrated origins and orientations are the quantities used in the world-frame CoP transformation of eq. (2.6).

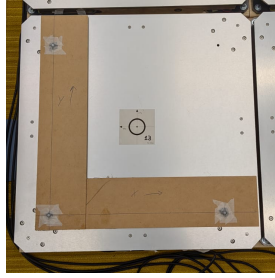


Figure 3.1: Custom L-shaped MoCap target used to register the position and orientation of each force-plate sensor in the common world frame $\mathcal{I}$.

3.1.2 Rigid-body marker frames, projected CoM references, and reference weights

Several retroreflective markers were fixed to the robots and objects in arbitrary, non-coplanar configurations. For a rigid entity a , the marker geometry defines a repeatable local MoCap body frame. Let $\mathbf{h}_{a,k}$ denote the planar world-frame position of its marker-frame origin and let $\theta_{a,k}^{\text{MC}}$ denote its measured yaw at sample k .

The marker-frame origin is generally not coincident with the vertical projection of the entity CoM. A separate static reference experiment was therefore performed on a six-degree-of-freedom Bertec force plate [23]. The Bertec plate provides a substantially more accurate reference CoP than the uniaxial floor plates. Its position in the MoCap volume was registered using its own proprietary calibration target. For a rigid entity a , let $\mathbf{c}_{a,k}^{\text{B}}$ denote the Bertec CoP expressed in $\mathcal{I}$. Under static equilibrium, the resultant CoP provides the vertical projection of the CoM. The constant body-frame offset $\mathbf{o}_a$ from the MoCap marker-frame origin to the projected CoM is estimated by

$$\mathbf{o}_a = \arg \min_{\mathbf{o}} \sum_{k \in \mathcal{K}_a^{\text{B}}} \|\mathbf{c}_{a,k}^{\text{B}} - (\mathbf{h}_{a,k} + \mathbf{R}(\theta_{a,k}^{\text{MC}}) \mathbf{o})\|_2^2, \quad (3.3)$$

where $\mathcal{K}_a^{\text{B}}$ is the set of valid samples in the Bertec reference recording. The projected CoM reference used in the subsequent CoP and tracking experiments is therefore

$$\mathbf{c}_{a,k} = \mathbf{h}_{a,k} + \mathbf{R}(\theta_{a,k}^{\text{MC}}) \mathbf{o}_a. \quad (3.4)$$

The same Bertec recordings provide an independent ground-truth weight. Let $F_{z,a,k}^{\text{B}}$ denote the vertical force measured by the Bertec plate. The reference weight of

entity a is

$$F_a^B = \frac{1}{|\mathcal{K}_a^B|} \sum_{k \in \mathcal{K}_a^B} F_{z,a,k}^B. \quad (3.5)$$

The equivalent mass is $m_a^B = F_a^B / g_0$. Thus, the Bertec plate measures the reference vertical force from which the corresponding equivalent mass is obtained. This quantity is the external ground-truth reference that the tracker-estimated neutral weight F_a^0 attempts to match. Separate static Bertec recordings were obtained for both human participants so that all weight errors are evaluated against the appropriate participant-specific reference.

Table 3.1: Bertec reference weights and corresponding equivalent masses.

Entity	Reference weight (N)	Equivalent mass (kg)
Vstone MecanumRover 3	230.14	23.47
Kettlebell	78.44	8.00
SoftBank Pepper	286.51	29.22
Bucket	148.13	15.11
Adult male 1	713.23	72.73
Adult male 2	812.26	82.83

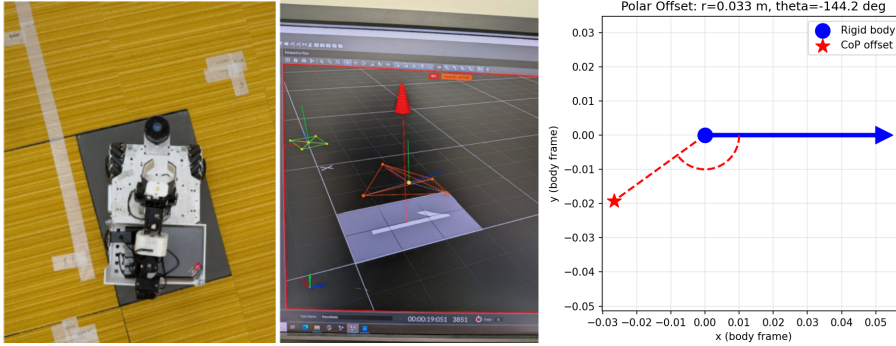


Figure 3.2: Rigid-body reference construction. The MoCap marker frame is related to the projected CoM by the fitted offset $\mathbf{o}_a$.

3.2 Single-plate CoP characterization

The registered MoCap reference was used to characterize the positional accuracy of the center of pressure (CoP) reported by an individual force-plate sensor. A rigid reference object was placed successively at a regular 5×5 set of static locations across the plate. The object's marker-frame-to-CoM offset was obtained in the Bertec experiment described in section 3.1.2. Consequently, the position reference in this experiment is the projected CoM defined by eq. (3.4), rather than the marker-frame origin.

Let $\mathcal{T}$ denote the set of tested plate-object combinations. For each $t \in \mathcal{T}$, let $s(t)$ and $a(t)$ denote the plate sensor and reference object, respectively. Let $\mathcal{I}_t$ be the set of valid test locations, and let $\mathcal{K}_{t,i}^{\text{FP}}$ and $\mathcal{K}_{t,i}^{\text{MC}}$ be the plate and MoCap sample sets from the static segment at location $i \in \mathcal{I}_t$. The mean CoP and projected-CoM positions are

$$\bar{\mathbf{r}}_{t,i} = \frac{1}{|\mathcal{K}_{t,i}^{\text{FP}}|} \sum_{k \in \mathcal{K}_{t,i}^{\text{FP}}} \mathbf{r}_{s(t),k}, \quad \bar{\mathbf{c}}_{t,i} = \frac{1}{|\mathcal{K}_{t,i}^{\text{MC}}|} \sum_{k \in \mathcal{K}_{t,i}^{\text{MC}}} \mathbf{c}_{a(t),k},$$

where $\mathbf{r}_{s(t),k}$ is the world-frame CoP reported by the plate and $\mathbf{c}_{a(t),k}$ is the projected-CoM reference. The position-error vector and its magnitude at location i are

$$\mathbf{e}_{t,i} = \bar{\mathbf{r}}_{t,i} - \bar{\mathbf{c}}_{t,i} = \begin{bmatrix} e_{t,i}^x \\ e_{t,i}^y \end{bmatrix}, \quad e_{t,i} = \|\mathbf{e}_{t,i}\|_2. \quad (3.6)$$

Let $N = \sum_{t \in \mathcal{T}} |\mathcal{I}_t|$ be the number of evaluated plate-reference pairs. The mean position error and position RMSE are

$$\bar{e} = \frac{1}{N} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} e_{t,i}, \quad e_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} e_{t,i}^2}.$$

For $q \in \{x, y\}$, the coordinate-wise spatial residual standard deviation is

$$\sigma_q^{\text{err}} = \sqrt{\frac{1}{N-1} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} (e_{t,i}^q - \bar{e}^q)^2}, \quad \bar{e}^q = \frac{1}{N} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} e_{t,i}^q.$$

Short-term CoP noise was evaluated separately from the spatial error. Let the number of extracted plate samples be

$$M = \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} |\mathcal{K}_{t,i}^{\text{FP}}|.$$

For $q \in \{x, y\}$, the pooled within-location standard deviation is

$$\sigma_q^{\text{noise}} = \sqrt{\frac{1}{M-1} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} \sum_{k \in \mathcal{K}_{t,i}^{\text{FP}}} \left(r_{s(t),k}^q - \bar{r}_{t,i}^q \right)^2},$$

where $\bar{r}_{t,i}^q$ is the q -coordinate of $\bar{\mathbf{r}}_{t,i}$. This measures frame-to-frame variation around each static location; it does not include the displacement between the mean CoP and its MoCap reference.

For each plate-object combination, the largest and smallest errors over the evaluated locations are

$$e_t^{\text{max}} = \max_{i \in \mathcal{I}_t} e_{t,i}, \quad e_t^{\text{min}} = \min_{i \in \mathcal{I}_t} e_{t,i}.$$

Their reported values are averaged over the tested combinations:

$$\bar{e}^{\text{max}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} e_t^{\text{max}}, \quad \bar{e}^{\text{min}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} e_t^{\text{min}}.$$

To quantify the spatial growth of position error, a zero-intercept radial model was fitted independently for each plate-object combination. Its center is expressed in the world frame and constrained to the rectangular extent of the MoCap reference centers,

$$\mathcal{B}_t = \left[\min_{i \in \mathcal{I}_t} \bar{c}_{t,i}^x, \max_{i \in \mathcal{I}_t} \bar{c}_{t,i}^x \right] \times \left[\min_{i \in \mathcal{I}_t} \bar{c}_{t,i}^y, \max_{i \in \mathcal{I}_t} \bar{c}_{t,i}^y \right].$$

For a candidate center $\mathbf{o} \in \mathcal{B}_t$, its distance to reference location i is

$$d_{t,i}(\mathbf{o}) = \|\bar{\mathbf{c}}_{t,i} - \mathbf{o}\|_2.$$

The model is

$$e_{t,i} = \beta_t d_{t,i}(\mathbf{o}_t) + \xi_{t,i}, \quad \beta_t \geq 0,$$

where $\xi_{t,i}$ is the residual. The fitted center and nonnegative slope are obtained jointly by bounded nonlinear least squares,

$$\left(\hat{\mathbf{o}}_t, \hat{\beta}_t \right) = \arg \min_{\substack{\mathbf{o} \in \mathcal{B}_t, \\ \beta \geq 0}} \sum_{i \in \mathcal{I}_t} [e_{t,i} - \beta \|\bar{\mathbf{c}}_{t,i} - \mathbf{o}\|_2]^2. \quad (3.7)$$

The optimization is started at the reference location with the lowest observed error, the centroid of all reference locations, and the four corners of $\mathcal{B}_t$. Each start uses the nonnegative zero-intercept least-squares slope for its initial center, and the solution with the smallest residual sum of squares is retained. A radial fit is reported only when at least three locations are available and the reference locations have nonzero extent along both coordinate axes. The reported radial error increase is the mean fitted slope,

$$\bar{\beta} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \hat{\beta}_t.$$

The aggregate results are summarized in table 3.2.

Table 3.2: Single-plate CoP position-error characterization against the offset-corrected MoCap projected-CoM reference.

Metric	Value	Unit
Mean position error	0.0210	m
Position RMSE	0.0232	m
Mean maximum position error	0.0382	m
Mean minimum position error	0.0036	m
Measurement-noise σ_x^{noise}	0.0022	m
Measurement-noise σ_y^{noise}	0.0020	m
Spatial residual σ_x^{err}	0.0126	m
Spatial residual σ_y^{err}	0.0140	m
Mean radial error gradient $\bar{\beta}$	0.1026	m/m

The mean CoP position error was 2.10 cm, whereas the within-location measurement-noise standard deviations were 2.2 mm and 2.0 mm along the x - and y -axes, respectively. The larger spatial residuals therefore reflect a position-dependent bias across the plate rather than short-term measurement noise. Across plate-object combinations, the fitted radial gradient was 0.1026 m/m, corresponding to an average predicted increase of 1.03 cm per 10 cm from the fitted zero-error center. The mean minimum and maximum errors were 0.36 cm and 3.82 cm, respectively. CoP error also appeared to decrease as the applied load increased, while measurements below approximately 40 N were unreliable; the available data were not sufficient to quantify this dependence.

The fitted zero-error center differed between plates and changed after the force-floor arrangement was physically adjusted. This spatial bias is therefore configuration dependent and cannot be represented by one global CoP offset.

The measured coordinate-wise spatial residual standard deviations were

$$\sigma_x^{\text{err}} = 0.0126 \text{ m}, \quad \sigma_y^{\text{err}} = 0.0140 \text{ m}.$$

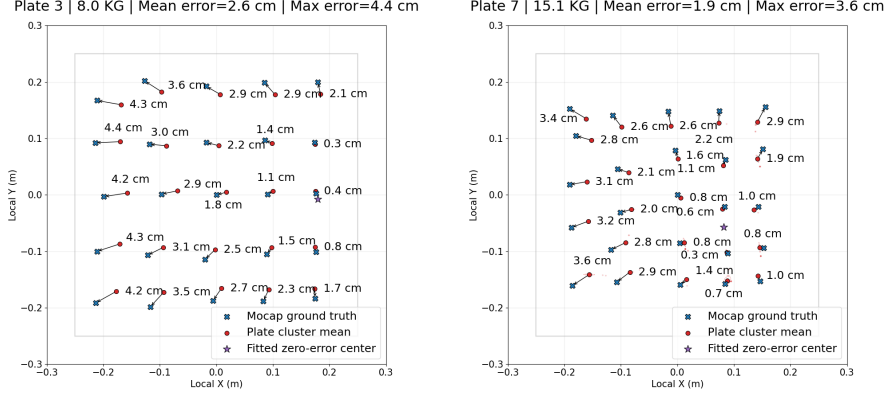


Figure 3.3: Single-plate CoP characterization of an 8 kg kettlebell on plate 3 (left) and a 15.1 kg bucket on plate 7 (right), evaluated against their offset-corrected projected-CoM references.

When cross-correlation is neglected, the CoP position error can be modeled as an isotropic covariance

$$\sigma_{\text{pos}} = \sqrt{\frac{(\sigma_x^{\text{err}})^2 + (\sigma_y^{\text{err}})^2}{2}} \approx 0.0133 \text{ m}, \quad \Sigma_{\text{pos}} \approx \sigma_{\text{pos}}^2 \mathbf{I}_2, \quad (3.8)$$

where $\mathbf{I}_2$ is the 2×2 identity matrix. The tracking model therefore uses the spatial residual covariance, whose scale includes the observed location-dependent bias, rather than the much smaller within-location measurement-noise covariance.

3.3 Solo and multi-agent tracking evaluation

A direct comparison with previously published tracking systems would not provide a meaningful baseline for this experiment. Camera-based, wearable-based, and alternative smart-floor systems observe different physical quantities and operate under different sensing assumptions, whereas the proposed method estimates persistent agent identities and force ownership using only the coarse vertical-force measurements of the instrumented floor. The appropriate baseline is therefore the same tracker operating under the simpler single-agent condition. These solo episodes characterize the position and weight-estimation behavior obtained without inter-agent force competition, while the multi-agent episodes quantify the additional degradation introduced by shared footprints, ambiguous ownership, and identity persistence.

3.3.1 Episode design and time synchronization

The evaluation comprises 82 one-minute episodes. Forty-one episodes are solo recordings in which one robot or one human traverses the instrumented floor using varied displacement patterns. The remaining 41 episodes are multi-agent scenes containing multiple robots, humans, and objects simultaneously. Both human participants are represented throughout the experiment. Their episode-level statistics are computed using their individual reference quantities and subsequently pooled for the reported human results. For the robots and objects, the same marker configuration from section 3.1.2 is used to define their MoCap reference frames. For the human participants, markers were placed bilaterally on the anterior superior

iliac spine (ASIS) and posterior superior iliac spine (PSIS), as well as on the hallux (HLX), fifth metatarsal head (HM5), and heel (CALD). Each force-floor recording is synchronized with its corresponding MoCap recording using a deliberate impact on a selected plate with a tracked marker. The impact appears as a sudden deceleration in the MoCap marker trajectory and a sharp peak in the measured vertical force. The two recordings are aligned at this event. Each force-floor sample is then associated with the temporally corresponding MoCap sample. Each tracker identity is associated with its corresponding MoCap entity when the agent is spawned, and this association remains fixed throughout the episode. Consequently, identity switches remain visible in the validation statistics rather than being removed through frame-wise reassociation.

3.3.2 Solo CoP distributions

Rigid robots. For a rigid robot a , the tracker position $\mathbf{r}_{a,k}$ is the position component of the persistent agent state defined in section 2.3.3. The robot-local frame is centered at the marker-frame origin introduced in section 3.1.2:

$$\tilde{\mathbf{r}}_{a,k} = \mathbf{R}(\theta_{a,k}^{\text{MC}})^{\top} (\mathbf{r}_{a,k} - \mathbf{h}_{a,k}). \quad (3.9)$$

The marker-frame-to-CoM offset identified in section 3.1.2 is used to express the ground-truth CoM within this local frame. Changes in wheel loading shift the resultant CoP relative to this CoM position, and the position-dependent plate error characterized in section 3.2 adds spatial bias. Each robot therefore produces a characteristic local CoP distribution in its marker-defined frame.

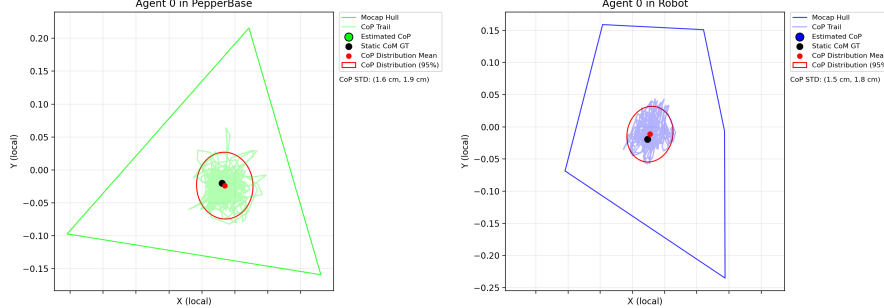


Figure 3.4: Example solo CoP distributions for Pepper and MecanumRover 3 expressed in their MoCap frames. The origin corresponds to the geometric center of the MoCap hull.

Human canonical support frame. A walking human does not provide a rigid planar support frame that remains fixed relative to the instantaneous CoP. The human tracker position $\mathbf{r}_{a,k}$ is therefore mapped into a canonical two-foot support frame before its distribution is evaluated. This transformation removes global translation, heading, and instantaneous stride geometry while retaining the recovered support position relative to the two feet. For foot $j \in \{R, L\}$ at sample k , let $\mathbf{r}_{j,k}^{\text{HLX}}$, $\mathbf{r}_{j,k}^{\text{HM5}}$, and $\mathbf{r}_{j,k}^{\text{CALD}}$ denote the hallux, fifth-metatarsal-head, and heel marker positions, respectively. Define the forefoot point and geometric foot center as

$$\mathbf{r}_{j,k}^{\text{toe}} = \frac{\mathbf{r}_{j,k}^{\text{HLX}} + \mathbf{r}_{j,k}^{\text{HM5}}}{2}, \quad \mathbf{r}_{j,k}^{\text{foot}} = \frac{\mathbf{r}_{j,k}^{\text{HLX}} + \mathbf{r}_{j,k}^{\text{HM5}} + \mathbf{r}_{j,k}^{\text{CALD}}}{3}.$$

The individual-foot and combined two-foot support polygons are

$$\mathcal{H}_{j,k}^{\text{foot}} = \text{conv} \{ \mathbf{r}_{j,k}^{\text{HLX}}, \mathbf{r}_{j,k}^{\text{HM5}}, \mathbf{r}_{j,k}^{\text{CALD}} \}, \quad \mathcal{H}_k = \text{conv} (\mathcal{H}_{R,k}^{\text{foot}} \cup \mathcal{H}_{L,k}^{\text{foot}}).$$

Thus, $\mathcal{H}_k$ represents the support area occupied by and between the two feet. Define

$$\begin{aligned} \mathbf{f}_{j,k} &= \mathbf{r}_{j,k}^{\text{toe}} - \mathbf{r}_{j,k}^{\text{CALD}}, \quad L_{j,k} = \|\mathbf{f}_{j,k}\|_2, \quad \mathbf{e}_{f,j,k} = \frac{\mathbf{f}_{j,k}}{L_{j,k}}, \\ \mathbf{e}_{r,j,k} &= \begin{bmatrix} e_{f,j,k,y} \\ -e_{f,j,k,x} \end{bmatrix}, \quad \mathbf{B}_{j,k} = [\mathbf{e}_{r,j,k} \quad \mathbf{e}_{f,j,k}]. \end{aligned} \quad (3.10)$$

Let L_c denote the fixed heel-to-toe foot length of the participant associated with agent a , and let $d = 0.5$ m be the canonical separation between the two foot centers. The transformation associated with foot j is

$$\begin{aligned} T_{j,k}(\mathbf{r}) &= \mathbf{r}_j^{\text{fix}} + \frac{L_c}{L_{j,k}} \mathbf{B}_{j,k}^\top (\mathbf{r} - \mathbf{r}_{j,k}^{\text{foot}}), \\ \mathbf{r}_R^{\text{fix}} &= \begin{bmatrix} d/2 \\ 0 \end{bmatrix}, \quad \mathbf{r}_L^{\text{fix}} = \begin{bmatrix} -d/2 \\ 0 \end{bmatrix}. \end{aligned} \quad (3.11)$$

Positions contained within an individual-foot support polygon use that foot's transformation directly. All other positions are transformed using inverse-square blending,

$$\tilde{\mathbf{r}}_{a,k} = \begin{cases} T_{R,k}(\mathbf{r}_{a,k}), & \mathbf{r}_{a,k} \in \mathcal{H}_{R,k}^{\text{foot}}, \\ T_{L,k}(\mathbf{r}_{a,k}), & \mathbf{r}_{a,k} \in \mathcal{H}_{L,k}^{\text{foot}}, \\ \frac{w_{R,k}T_{R,k}(\mathbf{r}_{a,k}) + w_{L,k}T_{L,k}(\mathbf{r}_{a,k})}{w_{R,k} + w_{L,k}}, & \text{otherwise,} \end{cases} \quad (3.12)$$

where

$$w_{j,k} = (\text{dist}(\mathbf{r}_{a,k}, \mathcal{H}_{j,k}^{\text{foot}}) + \varepsilon)^{-2}, \quad \varepsilon > 0,$$

and ε is a small numerical regularization constant. The resulting canonical representation preserves left-right and heel-toe support motion while removing the global walking trajectory. The baseline human CoP distribution is therefore evaluated in this canonical two-foot support frame.

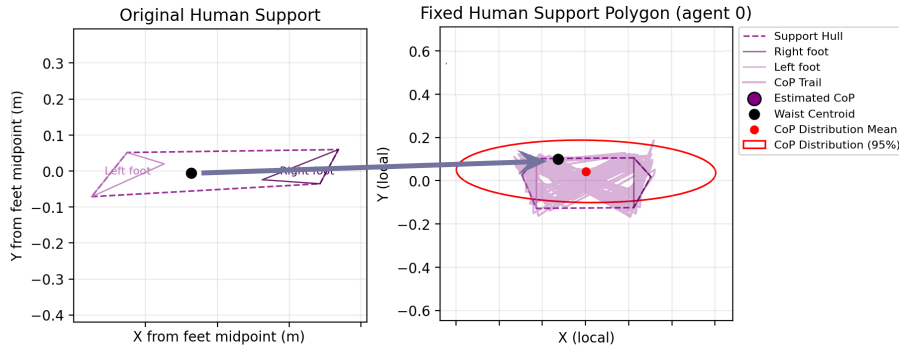


Figure 3.5: Example transformation of a human walking sample from the measured MoCap foot geometry into the canonical two-foot support frame used to aggregate human CoP distributions.

3.3.3 Episode-level evaluation metrics

For every episode e of agent a , the robot-local or human-canonical CoP samples are summarized after the episode has completed. Let $\mathcal{K}_{a,e}$ denote the set of valid samples and $n_{a,e} = |\mathcal{K}_{a,e}|$. The episode-mean transformed CoP is

$$\bar{\mathbf{r}}_{a,e} = \frac{1}{n_{a,e}} \sum_{k \in \mathcal{K}_{a,e}} \tilde{\mathbf{r}}_{a,k}, \quad (3.13)$$

and the coordinate-wise CoP standard deviation is

$$\sigma_{a,e,q} = \sqrt{\frac{1}{n_{a,e} - 1} \sum_{k \in \mathcal{K}_{a,e}} \left(\tilde{r}_{a,k,q} - \bar{r}_{a,e,q} \right)^2}, \quad q \in \{x, y\}. \quad (3.14)$$

For condition $c \in \{\text{solo}, \text{group}\}$, let $\mathcal{E}_a^c$ denote the corresponding set of episodes for agent a . The reported CoP spread is

$$\bar{\sigma}_{a,c,q} = \frac{1}{|\mathcal{E}_a^c|} \sum_{e \in \mathcal{E}_a^c} \sigma_{a,e,q}, \quad q \in \{x, y\}, \quad (3.15)$$

and its change between conditions is

$$\Delta\sigma_{a,q} = \bar{\sigma}_{a,\text{group},q} - \bar{\sigma}_{a,\text{solo},q}. \quad (3.16)$$

Weight estimation accuracy. The tracker maintains the estimated neutral weight F_a^0 defined in section 2.3.3 and updated following eq. (2.21). Weight accuracy is evaluated from the value reached by this estimate at the end of each episode. Let $F_{a,e}^0$ denote the final neutral-weight estimate for agent a in episode e . The independent reference weight F_a^B is obtained from the Bertec force-plate measurements described in section 3.1.2. The episode-level neutral-weight error is therefore

$$E_{a,e}^{\text{weight}} = |F_{a,e}^0 - F_a^B|. \quad (3.17)$$

The reported error for condition c is

$$E_{a,c}^{\text{weight}} = \frac{1}{|\mathcal{E}_a^c|} \sum_{e \in \mathcal{E}_a^c} E_{a,e}^{\text{weight}}. \quad (3.18)$$

This metric therefore evaluates the accuracy of the persistent neutral-weight estimate produced by the tracker at the end of each episode.

Mean CoP-to-CoM offset. For each rigid robot, eq. (3.9) expresses the recovered CoP in the local MoCap marker frame, whose origin is the geometric center of the marker set. The marker-frame-to-CoM offset identified in section 3.1.2 specifies the projected CoM with respect to this origin as $\mathbf{o}_a$.

Thus, the episode-mean transformed CoP from eq. (3.13) is measured relative to the marker-frame origin, and its difference from $\mathbf{o}_a$ is the mean CoP-to-CoM offset. For episode e , this offset is

$$E_{a,e}^{\text{CoP-CoM}} = \left\| \bar{\mathbf{r}}_{a,e} - \mathbf{o}_a \right\|_2, \quad (3.19)$$

and the reported offset for condition c is

$$E_{a,c}^{\text{CoP-CoM}} = \frac{1}{|\mathcal{E}_a^c|} \sum_{e \in \mathcal{E}_a^c} E_{a,e}^{\text{CoP-CoM}}. \quad (3.20)$$

This quantity includes both physical displacement of the resultant robot CoP from its CoM and tracker-induced bias. Its change between solo and group conditions measures the additional CoP-to-CoM displacement associated with multi-agent tracking. In contrast, $\sigma_{a,e,x}$ and $\sigma_{a,e,y}$ describe the spread of recovered CoP positions around their episode mean. No analogous CoP-to-CoM offset is defined for the human canonical representation.

Pooled human statistics. For the two human participants h_1 and h_2 , all episode-level quantities are calculated separately before aggregation. Let

$$\mathcal{E}_H^c = \mathcal{E}_{h_1}^c \cup \mathcal{E}_{h_2}^c$$

denote the combined human episode set for condition c . The reported human CoP statistics are obtained by pooling the episode-level values from both participants. Neutral-weight errors are pooled in the same manner, but the terminal estimate $F_{a,e}^0$ from each episode is compared with the Bertec reference weight of the participant who generated that episode. The two participants are therefore combined statistically without assigning them a common body weight.

Penalization of identity switches. CoP spread is particularly informative in the multi-agent condition because it is highly sensitive to identity switches. Under correct tracking, the transformed CoP remains confined to the relatively small support region around the CoM or canonical support region. If an identity switch occurs, however, the tracker continues to be evaluated in the original agent’s local frame while its CoP follows a different physical agent. The transformed CoP is then displaced by the separation between the two agents and can remain far outside the nominal local support region for the remainder of the episode. As a result, a distribution that would normally be compact around the origin can expand across a large portion of the workspace, causing a substantial increase in its standard deviation.

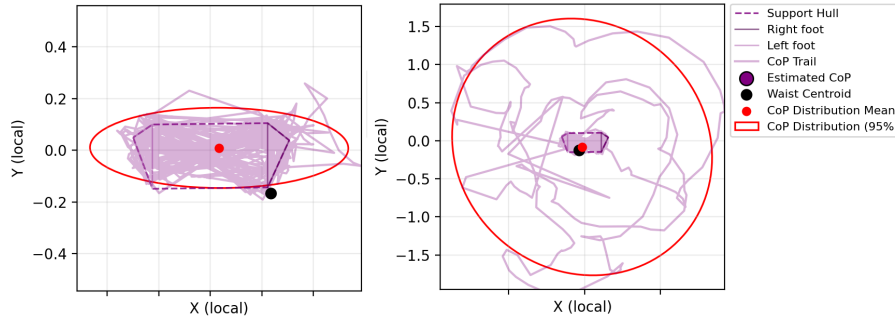


Figure 3.6: Example effect of an identity switch on the transformed CoP distribution. A correctly maintained identity (left) produces a consistent distribution, whereas an identity switch (right) displaces the transformed CoP by the separation between the two agents and substantially increases the standard deviation.

3.3.4 Results

The absolute changes from solo to multi-agent tracking are small in relation to the physical scale of the agents and their support distributions. For Pepper, the CoP standard deviation increases by 0.73 cm in x and 1.10 cm in y , while the neutral-weight error increases by 1.54 N. The MecanumRover 3 exhibits absolute CoP-spread

Table 3.3: MoCap validation over 82 one-minute episodes.

Feature	Agent	Solo avg.	Group avg.	Difference
CoP STD x (cm)	Pepper	1.77	2.50	+0.73
	MecanumRover 3	1.45	3.07	+1.62
	Humans (pooled)	24.37	24.98	+0.61
CoP STD y (cm)	Pepper	1.89	2.99	+1.10
	MecanumRover 3	1.43	3.41	+1.98
	Humans (pooled)	6.83	9.46	+2.63
Mean CoP–CoM offset (cm)	Pepper	0.80	1.38	+0.58
	MecanumRover 3	0.72	1.76	+1.04
	Humans (pooled)	–	–	–
Neutral-weight error (N)	Pepper	2.36	3.90	+1.54
	MecanumRover 3	1.84	3.05	+1.21
	Humans (pooled)	10.14	14.09	+3.95

increases of 1.62 cm in x and 1.98 cm in y , together with a 1.21 N increase in neutral-weight error. For the pooled human results, the CoP spread increases by 0.61 cm in x and 2.63 cm in y , while the neutral-weight error increases by 3.95 N. For the rigid robots, the mean CoP-to-CoM offset increases by 0.58 cm for Pepper and 1.04 cm for the MecanumRover 3. These changes show that multi-agent scenes introduce additional variation in the recovered CoP distributions and terminal neutral-weight estimates, but the resulting distributions remain close to their corresponding solo baselines in absolute terms. This behavior is observed across the different agent types despite their different geometries, support dynamics, and weights. These additional offsets remain comparable to the intrinsic spatial uncertainty of the individual force plates characterized in section 3.2.

Processing time. For the three-agent scenes, the mean processing time was 3.92 ms for the force solver, 2.70 ms for the position solver, and 2.85 ms for the remainder of the pipeline. The resulting mean total of 9.47 ms is below the 10 ms sensor period.

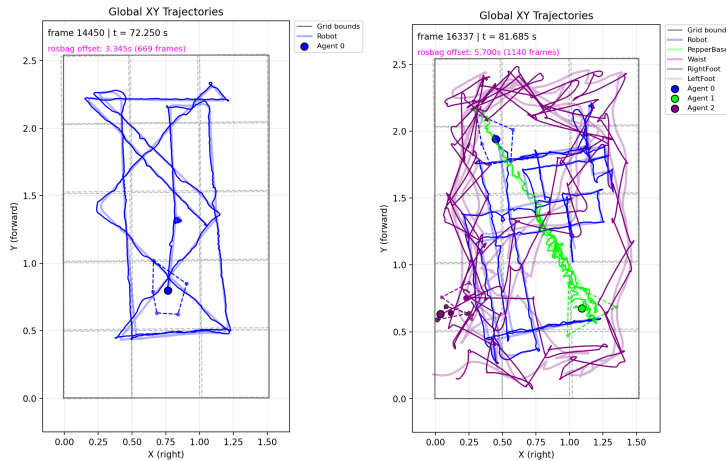


Figure 3.7: Example validation episodes for the single-agent baseline (left) and multi-agent tracking condition (right).

Chapter 4

Conclusion

4.1 Discussion

This thesis investigated whether a coarse grid of uniaxial force plates can provide useful gait- and form-factor-agnostic multi-agent tracking in shared human-robot environments. The experimental results demonstrate that, despite the limited spatial resolution and observability of the sensing hardware, the proposed system can recover persistent agent-level force and support information with sufficient accuracy and computational efficiency for real-time use.

The proposed system converts the total force, center of pressure (CoP), corner loads, and physical support of each active plate into persistent agent states. Unassigned footprint blobs provide evidence for agent birth, while established agents claim reachable measurements. The Force Share QP resolves fractional ownership of contested footprints, and the Sub-grid Position QP separates their contact positions while preserving the measured first moment. This combination recovers agent-level support below the 0.5 m plate pitch without violating the physical force and moment measured by the source plate. The recovered state also provides a basis for force-weighted influence fields, robot-model pose fitting, vertical touch reconstruction, and explicit pickup, carrying, and release relations.

The experiments first established the accuracy limits imposed by the sensing hardware itself. The single-plate mean CoP position error was 2.10 cm and the RMSE was 2.32 cm, despite within-location noise of only 2.0–2.2 mm. The dominant uncertainty is therefore a configuration-dependent spatial bias rather than short-term measurement noise. Its fitted magnitude increased on average by 1.03 cm per 10 cm from a plate-specific zero-error center, with mean minimum and maximum errors of 0.36 cm and 3.82 cm, respectively. This establishes a practical lower bound on the position accuracy that can reasonably be expected from tracking based on these measurements.

Against this hardware baseline, the tracker remained close to its solo behavior across the 41 multi-agent episodes. Relative to the 41 solo episodes, CoP spread increased by only 0.61–2.63 cm across agents and coordinate axes, while the rigid robots’ mean CoP-to-CoM offsets increased by 0.58–1.04 cm. Terminal neutral-weight errors increased by 1.21–3.95 N. These additional errors are small in absolute terms and remain comparable to the intrinsic spatial uncertainty of the individual force plates. The result is significant because the multi-agent condition adds the central ambiguities addressed by the proposed method, such as shared footprints, fractional force ownership, overlapping support regions, and persistent identity, while retaining approximately the same agent-level force and support information observed in the single-agent baseline. This behavior was obtained across Pepper, MecanumRover 3,

two human participants, and rigid objects with substantially different geometries, weights, and contact dynamics. The experiments therefore demonstrate that the proposed representation is practically usable as a gait- and form-factor-agnostic source of physical state in heterogeneous shared environments.

The system also satisfies the real-time requirement of the sensing platform. For the three-agent scenes, the Force Share QP required an average of 3.92 ms, the Sub-grid Position QP 2.70 ms, and the remainder of the tracking pipeline 2.85 ms, for a total mean processing time of 9.47 ms per frame. This corresponds to an average processing rate of approximately 105.7 Hz, exceeding the 100 Hz force-floor measurement rate. These results were obtained on a consumer laptop CPU without GPU acceleration, with the complete Python tracking pipeline executing in a single ROS 2 node without parallelization. Consequently, the current implementation already processes measurements faster than they are produced, while leaving substantial opportunities for further optimization and parallel execution.

The experiments do not imply that the force floor provides complete scene observability. Rather, they show that coarse vertical-force sensing can provide a distinct, computationally practical, and physically meaningful sensing channel for human-robot environments. Its principal strength is direct observation of support, load transfer, and contact, while its principal weakness remains identity and motion ambiguity when multiple physical explanations are consistent with the same vertical-force measurements.

4.2 Limitations

The main unresolved weakness is identity stability. The current solvers select the lowest-cost feasible explanation from the information available at each frame. When two agents produce physically equivalent or nearly equivalent force configurations, a locally preferable assignment can transfer measurements from one identity to another. Once such a switch enters the persistent state, the resulting history can reinforce the incorrect explanation in subsequent frames. Exact force and first-moment conservation prevent physically inconsistent solutions, but they cannot distinguish configurations that are observationally equivalent under the available vertical-force measurements. This is the same fundamental difficulty that motivates explicit merged-measurement handling in multi-target tracking [16, 17].

Identity management is incomplete not only when agents cross or merge, but also when they leave the observable floor. Agent birth is supported by explicit unassigned-force evidence, whereas disappearance is more ambiguous: an agent that stops receiving measurements may have left the floor, become temporarily unobservable, or had its support incorrectly assigned elsewhere. A robust tracker therefore requires an explicit exit and death model that combines proximity to the floor boundary, duration without compatible support, predicted motion, and competing ownership explanations. Without such a model, stale identities can persist after an agent has physically left the scene and can subsequently compete for measurements belonging to active agents.

A second limitation is that the principal ownership and position optimizations are solved as a sequence of individual-frame problems. Historical state enters through prediction and selected cost terms, but the optimization itself does not jointly enforce consistency over several frames. Consequently, a solution that is locally inexpensive but inconsistent with the longer trajectory can be accepted before later frames provide enough evidence to reveal the error. This is particularly problematic during crossings, prolonged shared support, rapid load transfer, and temporary loss of observable footprints.

The representation is also predominantly deterministic. Each frame produces a single

ownership and position estimate even when several explanations are nearly equally compatible with the measurements. The optimizer therefore collapses ambiguity into one solution rather than preserving uncertainty about agent position, force ownership, or identity. This can create overconfident state updates precisely in the configurations where the uniaxial floor provides the least information.

A further limitation concerns simultaneous agent entry. When multiple previously unseen agents enter the instrumented area while already in close contact or spatial overlap, their unassigned footprints may initially form a single connected blob. Because agent initialization relies on the separation and persistent tracking of these unassigned measurements, the system can represent them as a single agent until their footprints become sufficiently separated to support distinct tracks. Improving this case would require birth mechanisms capable of reasoning about multiple latent agents within an initially merged observation.

The sensing principle also assumes that measured ground-reaction forces remain sufficiently localized to the physical contacts that generated them. This assumption can break down when loads are transmitted through structures that redistribute pressure over a large area, such as a rigid floor layer placed above the force plates, furniture spanning several plates, or other mechanically coupled surfaces. In these cases, a localized contact may appear as a broad distributed load whose measured CoP and spatial support no longer correspond directly to the original contact location. Such environments would require an additional mechanical transfer model or calibration procedure and therefore fall outside the assumptions of the present system.

The floor should consequently be interpreted as a complementary sensing modality rather than a complete standalone perception system. It directly observes support and vertical load transfer, but identity remains weak during symmetric crossings, prolonged merged support, and other configurations in which several agents can explain the same resultant load. Purely horizontal contacts and body motion that do not alter ground support are also outside the direct observability of the uniaxial floor.

4.3 Future work

The first priority is stronger lifecycle and identity management. Identity resolution should operate over a longer horizon than the current frame-wise assignment and should explicitly model both ambiguous merges and agents leaving the instrumented area. Rather than immediately committing to one locally cheapest explanation, the tracker could preserve several plausible hypotheses through ambiguous intervals and resolve them once the trajectories separate or additional force evidence becomes available. Exit hypotheses should likewise accumulate evidence from boundary proximity, predicted motion, time since the last compatible support measurement, and the absence of plausible force ownership before an identity is removed. Learned neutral force, recurring gait patterns, robot support geometry, and characteristic load-transfer signatures could provide additional identity evidence without relaxing the exact per-plate force and moment constraints. Exploratory attempts to address these issues are illustrated in appendix A.1 and appendix A.2.

More generally, the Force Share and Sub-grid Position stages should be extended from single-frame optimization to temporally consistent estimation. A moving window or trajectory-level formulation could jointly optimize ownership and contact position over several consecutive frames, penalizing solutions that require implausible changes in identity, force share, or support location. Temporal consistency would allow future measurements to disambiguate earlier frames and should reduce the tendency of a single locally attractive assignment to become a persistent tracking

error.

A corresponding extension is to replace the deterministic state estimate with a probabilistic representation. Instead of returning only one ownership matrix and one contact-position solution, the tracker should represent uncertainty in agent state, footprint ownership, and sub-grid position. Ambiguous scenes could then produce broad or multimodal beliefs rather than artificially precise estimates. These uncertainties could be propagated through prediction, assignment, and agent-state updates, allowing downstream HRI reasoning to distinguish well-observed interactions from hypotheses supported by weak or ambiguous floor evidence.

The footprint model should also be learned online from each agent’s observation history. The persistent tracker already accumulates repeated examples of where an agent places force relative to its own state. These observations can be used to construct an adaptive agent-specific footprint distribution that captures characteristic support geometry and load placement. Assignment could then move from the current spread-based spatial cost to a probabilistic cost based on the likelihood of each candidate footprint under the predicted distribution. A footprint would therefore be favored not merely because it is close to the agent, but because its position and force are consistent with support patterns previously observed for that agent. Online adaptation would allow the model to specialize to different humans, robots, objects, postures, and changing contact patterns without requiring a fixed form-factor model. An initial exploration of this idea is presented in appendix A.3.

The final research direction is to evaluate the recovered physical state for long-horizon human–robot interaction rather than tracking alone. The force-weighted influence field introduced in this thesis provides a compact representation of where each agent is physically acting on its surroundings. These fields can be combined with the explicit carrying and contact relations into a dynamic interaction graph whose nodes represent agents and objects and whose edges represent spatial influence, support transfer, touch, carrying, and other physical relations. A temporal model, such as a Transformer operating on the evolving graph or its interaction tokens, could then detect and predict long-horizon HRI patterns from sequences of physical interactions rather than from isolated events. This would extend the floor from an agent tracker into a privacy-preserving physical interaction sensor capable of representing how people, robots, and objects influence one another over time. An initial exploration of this idea is presented in appendix A.4.

Together, these extensions would move the current system from deterministic, frame-local force decomposition toward temporally coherent probabilistic scene estimation and long-horizon interaction understanding. The present experiments establish the necessary first step: heterogeneous agents can already be tracked from coarse vertical-force measurements with multi-agent degradation close to the sensor’s intrinsic error and with processing fast enough to operate at the full sensor rate on modest CPU hardware.

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Appendix A

Exploratory tracking and recognition experiments

All of the methods described in this appendix were exploratory experiments performed during the research. They were not included in the main text because they were not developed to a stable state. The descriptions below therefore document the ideas that were investigated and the behaviour of the corresponding prototypes; they should not be read as validated parts of the final tracking and interaction-recognition pipeline.

A.1 Identity reassignment and switch recovery

The identity-reassignment experiment explored whether persistent identity switches could be corrected after they had already been accepted by the main tracking pipeline. Rather than operating at the full tracking frequency, the procedure was run periodically as a separate recovery stage. For a contested group of agents and footprints, it reconsidered the current distribution of footprint force among the agents and searched for an alternative interpretation.

Candidate redistributions were evaluated primarily according to two criteria: how closely the total force assigned to each agent matched its estimated neutral force, and how spatially compact the footprints assigned to that agent were. The solver therefore favored interpretations in which each agent retained a physically plausible weight while its supporting footprints remained concentrated around a common position. Because footprint ownership and agent position depend on one another, assignment and position were updated iteratively from several initial configurations. The best alternative was compared with the current tracked interpretation. If no lower-cost redistribution was found, the existing identities were retained. If a redistribution produced a sufficiently better force and spatial consistency, the corresponding footprint ownership was replaced and the affected agents were moved directly to the positions implied by their new footprint distributions. This discontinuous correction effectively allowed an agent to “teleport” across an erroneous identity switch rather than attempting to recover gradually through the normal motion model.

The approach remained experimental and was disabled by default. Its effectiveness depended strongly on the neutral forces being sufficiently different between agents and on the relative weighting of the force-mismatch and spatial-spread objectives. In particular, switches between agents of similar weight remained fundamentally ambiguous from the available force measurements alone.

A separate temporal safeguard explored whether a completed reciprocal switch could be recovered after more evidence had accumulated. For every eligible agent

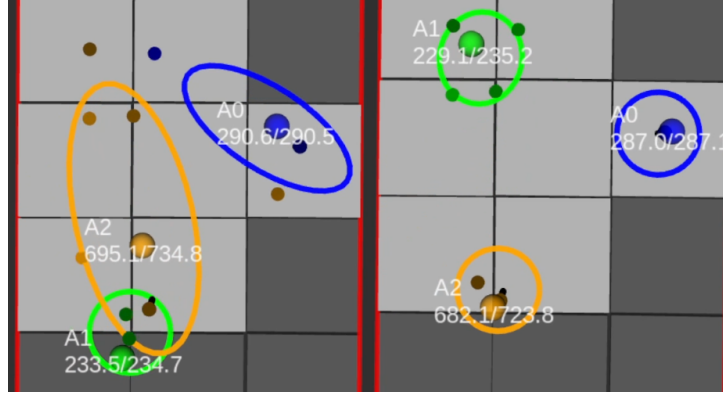


Figure A.1: Illustration of an identity switch between a human (yellow) and a MecanumRover 3 (green), before (left) and after (right) the reassignment. The solver reassigns the footprints to match the neutral forces of the two agents and minimize the spread, and moves them to the new centroid positions.

it retained the assigned force and neutral-force target over a rolling five-second window. The neutral targets were taken from the beginning of the window, before the motion being assessed. An agent became a candidate only when its average assigned force differed from its own earlier target by more than 10%. Two candidates were considered a reciprocal switch when each average was also within 10% of the other candidate's earlier target. In that case their footprint-assignment lists were exchanged while the identity-specific values, including the neutral forces, stayed with the identities. Their displayed positions were then snapped towards the new assignment centroids only when those centroids lay within the configured snap distance. The history was cleared after a recovery so that the same pre-switch evidence could not immediately reverse the change.

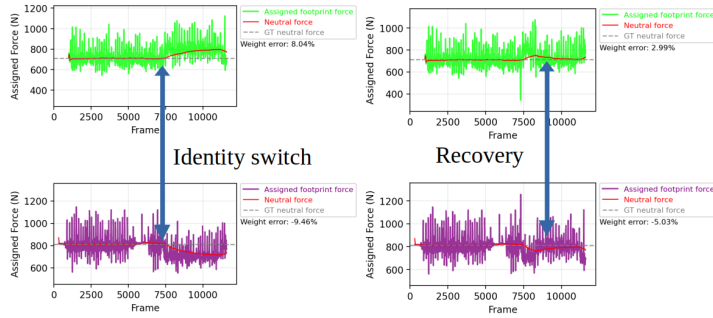


Figure A.2: Illustration of identity-switch recovery by neutral-weight matching. The temporal safeguard accepts a switch only when both identities cease to match their own earlier neutral forces and instead match one another's targets.

A.2 Agent removal solver

The Removal Solver considered the case in which an agent left the instrumented area while still belonging to a contested cluster. For agents with a low minimum-force bound, such as a human allowed to become temporarily unloaded during hopping or

stepping, disappearance could be difficult to distinguish from a brief reduction in support. Rather than removing the agent, the assignment solver would therefore continue to preserve it by allocating small amounts of force from nearby agents. This produced a persistent “parasite” agent that remained active by leeching off from the remaining agents, despite no longer having a physical counterpart on the instrumented floor.

Rather than asking a continuous optimizer to determine whether an agent was absent, a sustained absence of force first triggered the removal solver, using a criterion analogous to the sustained-excess mechanism in section 2.3.7. Once triggered, the solver generated a set of removal hypotheses by treating each agent in the contested cluster, in turn, as the potentially absent agent.

For each hypothesis, it copied the current footprint shares of all remaining agents and redistributed the removed agent’s share only to agents that were already supported on the same force plate. If several such agents were eligible, the share was divided equally between them. This same-plate rule was intended to avoid implausible transfers across disconnected supports and made every hypothesis deterministic.

Each hypothetical removal was judged in two ways. First, the force assigned to each remaining agent was compared with that agent’s neutral force. Second, the spread of the footprints around the resulting force-weighted agent position was penalized. The first term asked whether the remaining measured load could be explained by the agents that were still present; the second discouraged one agent from acquiring a widely separated collection of supports. The candidate with the lowest combined cost was returned together with the redistributed assignments and positions. Equal-cost cases were resolved by choosing the lowest agent identifier, making repeated runs reproducible.

This procedure was useful in simple scenarios with compact form factors and clear weight differences between agents like single-human, multiple robot scenes. It was not, however, a complete probabilistic absence detector.

A.3 Online fitting of local footprint distributions

The footprint-fitting experiment explored whether an agent’s accumulated local contact positions could be summarized by a structured probability density rather than by one fixed geometric footprint. Its input was an unordered set of two-dimensional contact positions expressed in the agent’s local frame.

Before enough observations were available, the model used an isotropic Gaussian prior centred on the local origin, with 95% of its probability mass inside a radius of 0.6 m by default. A DeepSets-style [36] neural network then read the contact positions one at a time, mean-pooled their learned encodings, and decoded a structured mixture. The mixture allowed up to four circular ring components and four full-covariance Gaussian lobes. Rings could describe contacts distributed around the body, whereas Gaussian lobes could describe localized or elongated support regions. The decoder predicted each component’s probability and geometric parameters while maintaining positive widths and valid Gaussian covariances.

The network was warm-started and refitted after every ten new samples using at most the latest 1000 positions. It maximized the likelihood of the observed positions while penalizing unnecessary components and strongly anisotropic Gaussian shapes. Components receiving less than 10% mean posterior support were gated off, nearby Gaussian lobes were merged, and samples belonging to a pruned component were absorbed into the strongest retained component. Minimum ring and Gaussian widths of 0.04 m prevented collapse onto individual points. For stability and smooth update, each newly evaluated density contributed only 20% of the result, with the previous density retaining the other 80%.

The qualitative fits in the two reserved panels below illustrate why a structured mixture was investigated. The ANYmal samples separated into two side regions, consistent with spatially distinct groups of leg contacts. The MecanumRover 3 samples instead formed a compact, approximately annular distribution around the local origin, consistent with repeated contacts surrounding the centre of the chassis. The contrast suggested that one isotropic footprint model would discard useful agent-specific structure.

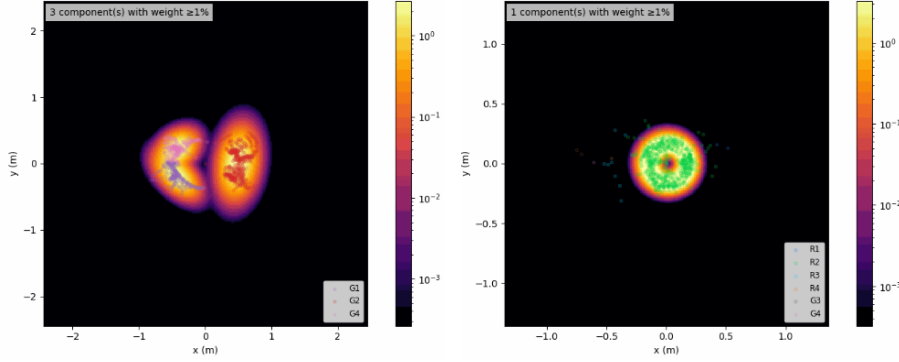


Figure A.3: Comparison of the footprint fits for ANYmal (left) and the MecanumRover 3 (right).

A.4 Directed graph representation for interaction recognition

The graph-neural-network experiment investigated whether the tracked agents and the relations between them could be used to recognize directed physical interactions. The recognition pipeline operated entirely downstream of tracking: it consumed the estimated agent states and force-derived relations, but did not feed any information back into the tracker or modify its force and position estimates.

At each tracker frame, every present agent was represented as a node in a directed graph. The node features described both the agent’s physical state and its recent dynamics, including neutral and measured vertical force, force rate and acceleration, position, velocity, acceleration, and the parameters of its spatial influence field. Dynamic force quantities were normalized by the agent’s neutral force so that agents with different body weights could be represented on a more comparable scale.

Each ordered pair of distinct agents defined a possible directed edge. Edge features described the relative state of the pair, including relative position, velocity and acceleration, Euclidean distance, whether the agents were approaching one another, a covariance-aware influence distance, and the overlap of their spatial influence fields. A measured edge was considered active when the pair was either sufficiently close or their influence fields overlapped beyond a configured threshold. Because the graph was directed, the source–target ordering was preserved even when some underlying geometric quantities, such as overlap, were symmetric.

The frame-level graph was then summarized at 20 Hz for interaction recognition. Each node summary contained the most recent position together with aggregated motion, force, transient-force, spatial-covariance, presence, and age information. Each ordered edge summary similarly captured recent relative geometry, proximity, accumulated influence-field overlap, force contrast between the two agents, and the fraction of frames for which the measured edge was active. These summaries

were stored in fixed-size tensors supporting up to six simultaneous agents, with 20 channels per node and 22 channels per ordered pair. A two-second temporal window therefore contained 40 recognition bins, while node and pair masks distinguished valid observations from padding.

Separate node and edge encoders (two-layer MLPs with SiLU activation) mapped these summarized features into learned latent representations. The encoders were first pretrained using self-supervision on the same 82 episodes used for the tracking experiments. Since the recorded scenes contained only a limited number of simultaneously present agents, segments from different episodes were artificially combined to construct denser synthetic scenes. This exposed the encoders to a wider range of multi-agent graph configurations without requiring additional physical recordings. The encoded graph was then processed by two edge-conditioned GraphSAGE layers [37]. For each target node, messages from all co-present source nodes were aggregated using both mean and maximum operations, while the corresponding edge embedding provided relational context.

After message passing, a representation was constructed for every ordered source–target pair. It combined the embeddings of both endpoint agents, their difference, their elementwise interaction, and the corresponding ordered edge embedding. Preserving this ordering was essential for directed recognition: exchanging the two agents produced a different pair representation and could therefore distinguish, for example, the initiator of an interaction from its receiver.

A causal temporal-convolution network [38] processed each ordered pair independently over the two-second history. Unlike the self-supervised encoder pretraining, the final interaction-recognition stage was trained on separately recorded and manually labeled interaction sequences. In the experimental setup, the target event was a directed *shove*, and the model used the temporal pair representation to identify the actor, the receiver, and the interaction class.

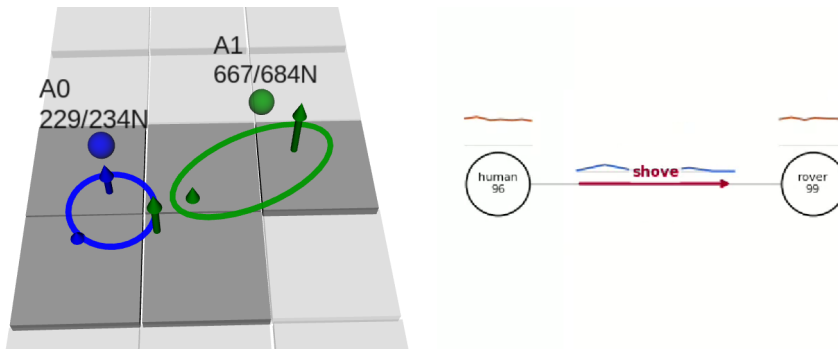


Figure A.4: Example directed graph representation and recognition of a human (green) shoving a MecanumRover 3 (blue).